

Faculty of Science and Technology,
UNIVERSITY OF MACAU

ECEN3003 INTRODUCTION TO BIOMEDICAL ENGINEERING

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Laboratory Project Report

Name & Student no.: Wu Yu Fei (D-C2-2886-2) & Ku Hou Ieong (D-C2-2789-2)

Faculty: Department of Electrical and Computer Engineering

Email: dc22886@umac.mo, dc22789@umac.mo

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I. INTRODUCTION

During this Laboratory session, we need to conduct 5 experiments covering what we have learnt in class about how to measure ECG signals using Einthoven limb leads (Experiment 1), how to measure EMG signals and analyze their relationship with muscle fatigue (Experiment 2), how to measure blood pressure using the palpation method and the Oscillometric method (Experiment 3), how to analyze the EMG signals under different muscle forces (Experiment 4), and finally to analyse and understand the circuit composition and structure of an instrumentation amplifier with three op-amps configuration. This process from classroom knowledge to hands-on experiments deepened our understanding.

The process and results of each experiment are recorded as follows.

II. LABORATORY #1

ECG (Electrocardiogram) measurement and analysis

A. Objectives

To conduct an ECG measurement and correlate it with the flow of blood.

B. Overview of principles and procedures

This experiment requires us to measure the ECG signal through the basic method of measuring Einthoven Limb Leads, which satisfies the relationships as follows: Lead I: $VI = \Phi_L - \Phi_R$; Lead II: $VII = \Phi_F - \Phi_R$; Lead III: $VIII = \Phi_F - \Phi_L$. By measuring the voltages of each lead that can objectively reflect the contraction and diastole of the heart in a specific direction, we can get the ECG signal. Then after some modulation, doctors will use the more obvious signals measured from lead1 or lead2 for diagnostic analysis. Also, as we learned in class, this kind of measurement of three leads can lead to many MIs being missed, and that 'silent areas' of the heart's electrical activity were not detectable. Therefore, the methods of uni-polar ECG measurement with Wilson central terminal (CT) and Goldberger Augmented leads measurement are more widely used nowadays.

According to the principle and the guidance, it's clear that the right leg should be always grounded as the reference, and the right arm, left arm and left leg should be connected to the positive or negative electrode accordingly as shown in Figure 1, depending on which lead is measured. After getting the voltages of two nodes, then subtracted to get the voltage value of the lead and measured with the blood flow trace got from the finger pulse transducer for comparison. Finally, we got the results shown below.

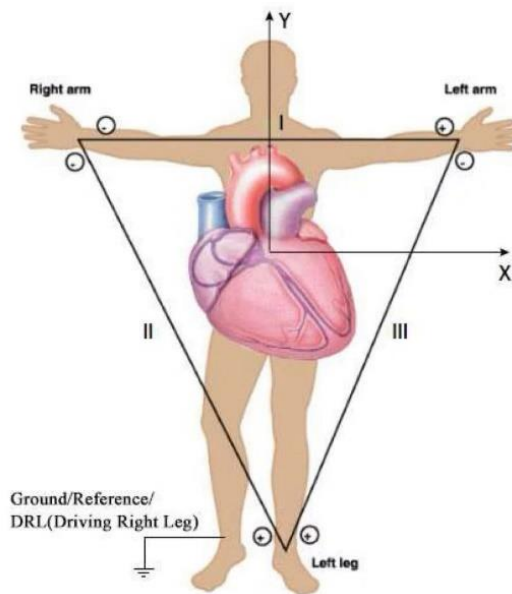


Fig.1. The connections of measurements.

C. Results

1. Show all the measurement results during the experiment.

Results obtained from volunteer 1 Ku Houieong:



Fig.2. The measurement results of the lead I signal (Channel 2: signal of finger pulse transducer; Channel 3: ΦL connected to the positive electrode; Channel 4: ΦR connected to the negative electrode; Channel 5: Lead I: $V_I = \Phi L - \Phi R$).

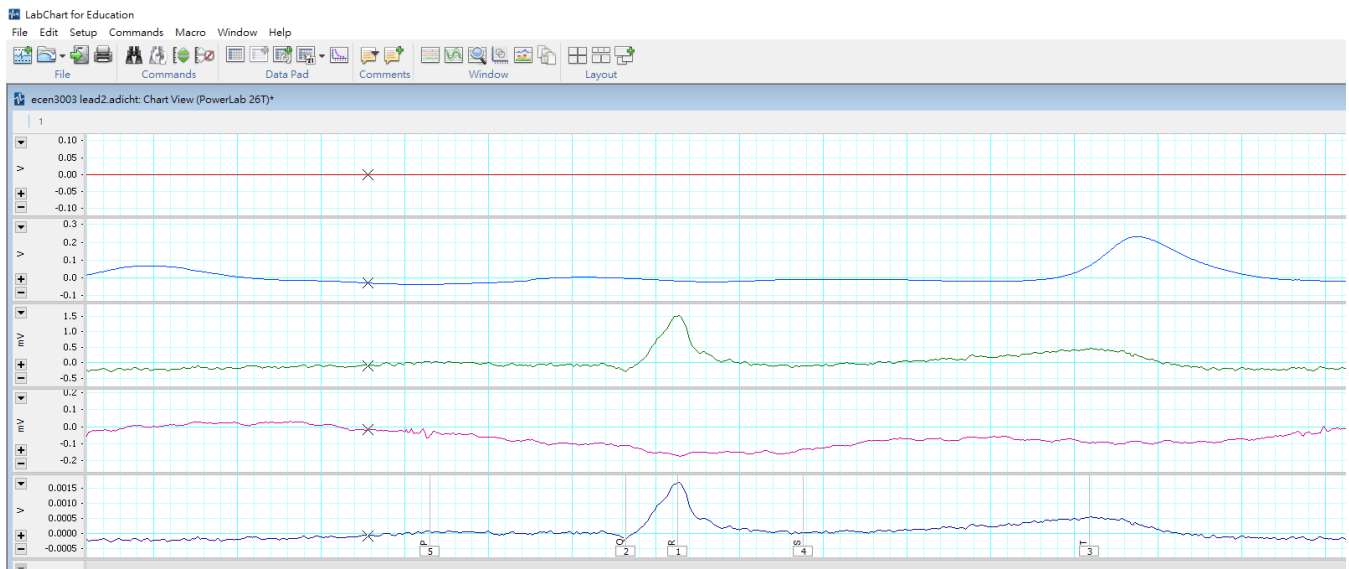


Fig.3. The measurement results of the lead II signal (Channel 2: signal of finger pulse transducer; Channel 3: ΦF connected to the positive electrode; Channel 4: ΦR connected to the negative electrode; Channel 5: Lead II: $V_{II} = \Phi F - \Phi R$).|



Fig.4. The measurement results of the lead III signal (Channel 2: signal of finger pulse transducer; Channel 3: ΦF connected to the positive electrode; Channel 4: ΦL connected to the negative electrode; Channel 5: Lead III: $V_{III} = \Phi F - \Phi L$).

Results obtained from volunteer 2 Wu Yufei:

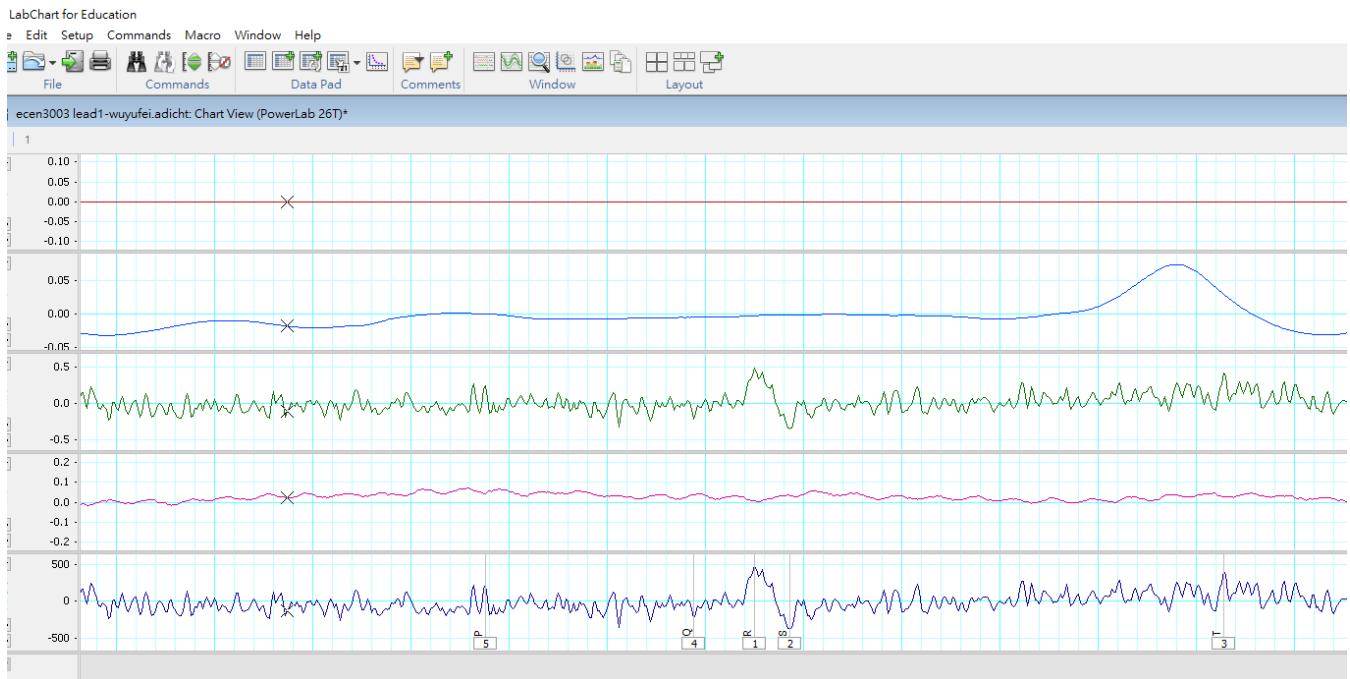


Fig.5. The measurement results of the lead I signal (Channel 2: signal of finger pulse transducer; Channel 3: ΦL connected to the positive electrode; Channel 4: ΦR connected to the negative electrode; Channel 5: Lead I: $VI = \Phi L - \Phi R$).

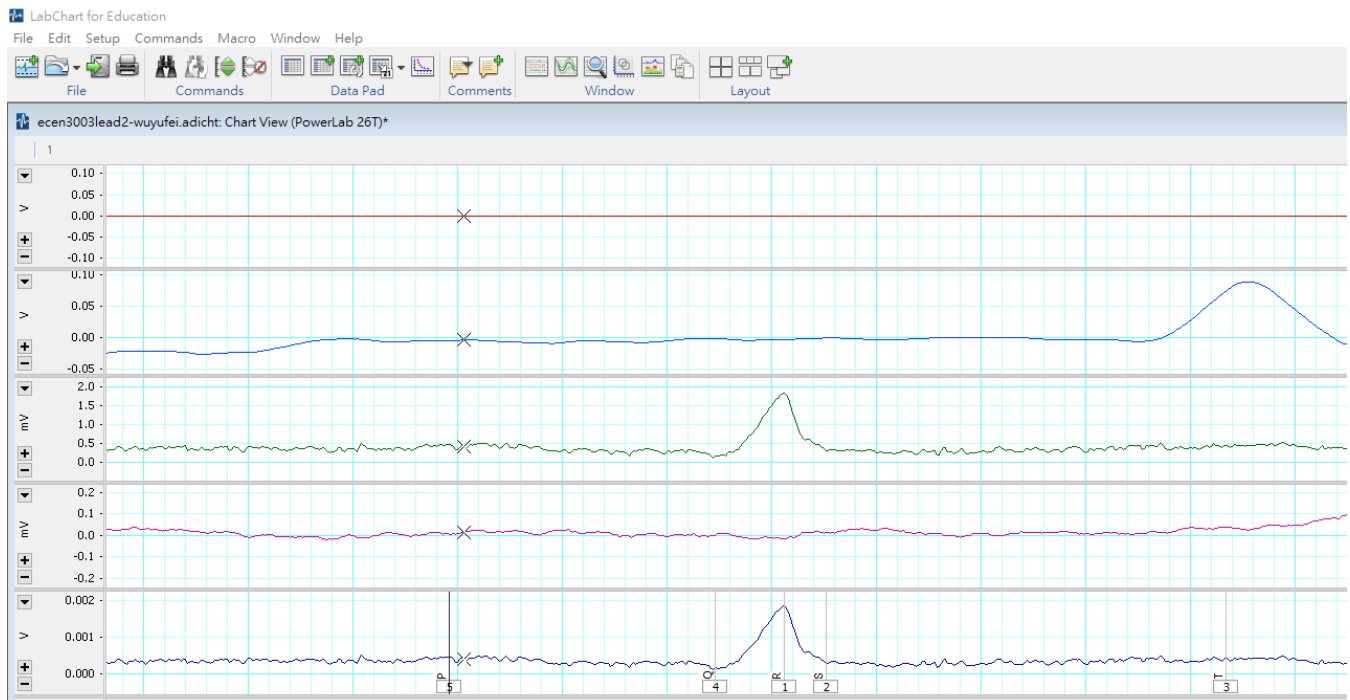


Fig.6. The measurement results of the lead II signal (Channel 2: signal of finger pulse transducer; Channel 3: ΦF connected to the positive electrode; Channel 4: ΦR connected to the negative electrode; Channel 5: Lead II: $VII = \Phi F - \Phi R$).

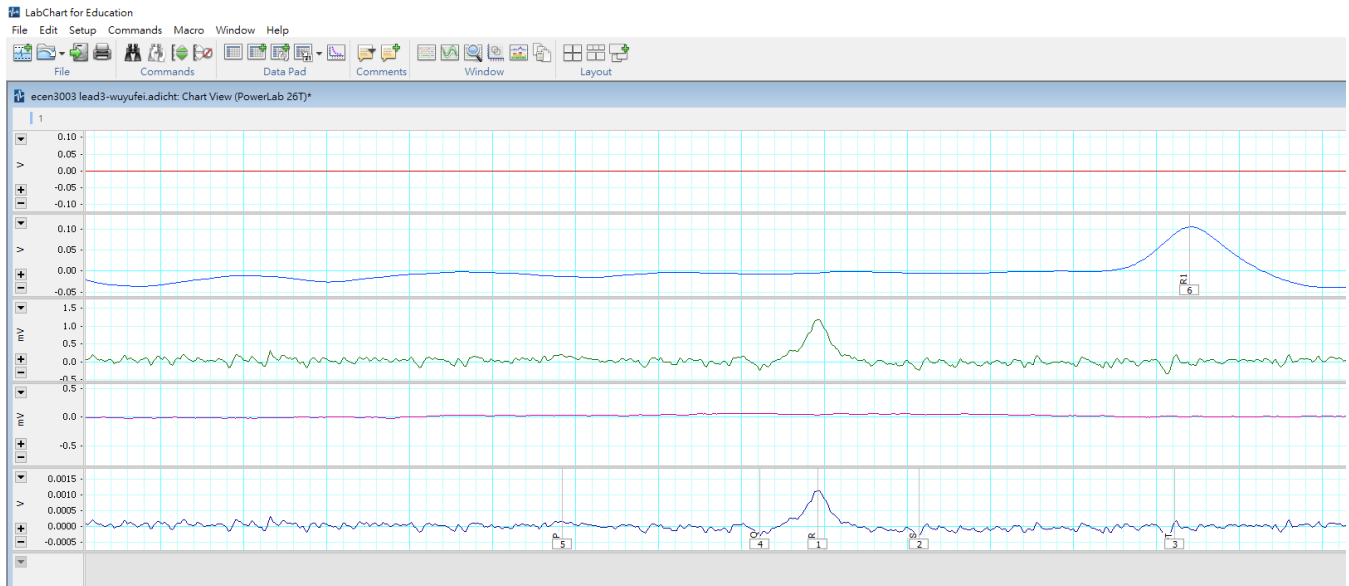


Fig.7. The measurement results of the lead III signal (Channel 2: signal of finger pulse transducer; Channel 3: ΦF connected to the positive electrode; Channel 4: ΦL connected to the negative electrode; Channel 5: Lead III: $V_{III} = \Phi F - \Phi L$).

2. What produces the QRS complex in the ECG?

As Figure 8 shows, the QRS complex in ECG signal mainly reflects the heart process of cardiac depolarization and ventricular contractility in different directions. The signal of the heart starts from the Sinoatrial node, and then atrial depolarization occurs at about 80 with the ECG signal starting to rise to the “P” point. After that, since the propagation through the AV junction is very slow, results in a delay in the progress of the activation. Next, the heart signal along the Purkinje fiber to the inner wall of the ventricles leads to the “Q” point of the ECG signal. Eventually, depolarization occurs on both sides of the septum, and early apical activation is also occurring, so the resultant vector points to the apex and because the left ventricular wall is thicker, activation of the left ventricular free wall continues even after depolarization of a large part of the right ventricle, leads the ECG signal rising till point “R” and then get down to the point “S”.

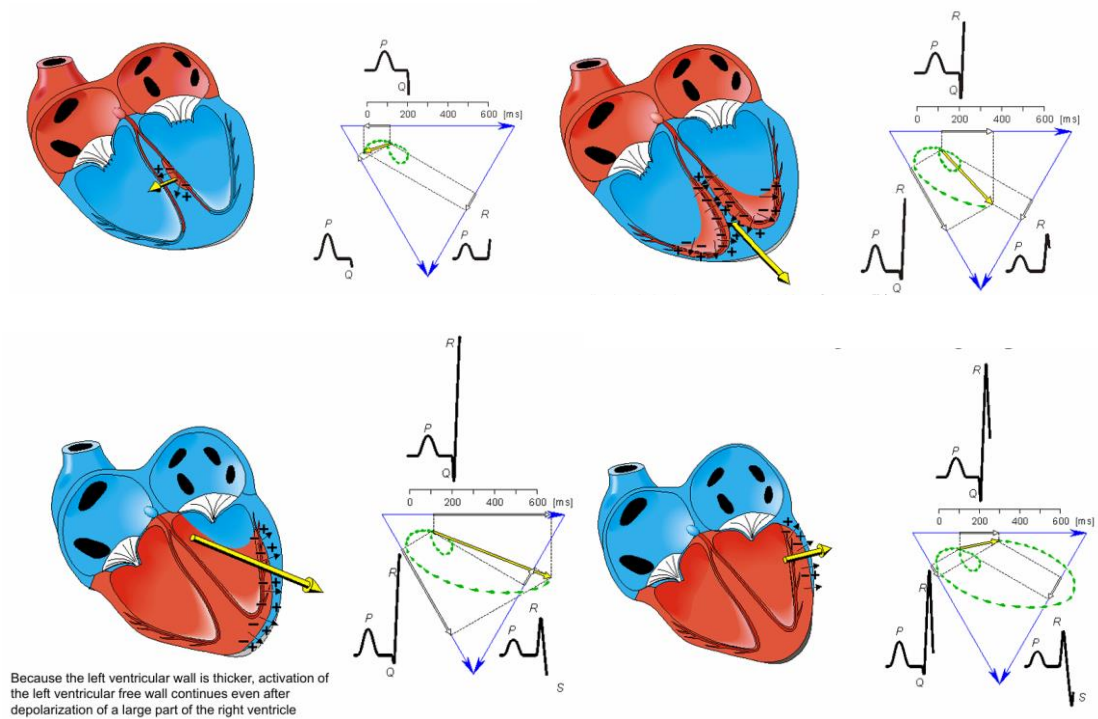


Fig.8. The formation of the ECG signal.

3. Please clearly label the QRS Complex in ECG for all recordings.

The label of the QRS complex is shown at Channel 5 in Figures of Q1. Since the signals of leadII and leadIII are not obvious, the measurement ranges of the y-axis is obtained in order to clearly mark them. Also, due to individual differences, the accuracy of ECG signals measured in different individuals will not be the same. As shown in the results, we can find that the ECG signals measured from volunteer 1 are more evident.

4. & 5. Calculate the Heart rate from the recorded ECG signal. & Calculate the Heart rate from the recorded Pulse wave.

Channel	Number	Time	Comment
5	5	4.55 s	P
5	4	4.649 s	Q
5	1	4.673 s	R
5	2	4.699 s	S
5	3	4.925 s	T
5	8	5.69 s	R2
2	6	9.06 s	R1
2	7	10.053 s	R2
All	9	10.699 s	New Comment

Channel	Number	Time	Comment
5	5	3.315 s	P
5	2	3.432 s	Q
5	1	3.463 s	R
5	4	3.538 s	S
5	3	3.709 s	T

Channel	Number	Time	Comment
5	5	6.542 s	P
5	2	6.656 s	Q
5	1	6.687 s	R
5	3	6.748 s	S
5	4	6.944 s	T

Fig.9. The information of the labeled QRS Complex of the ECG signal obtained from volunteer 1 Ku Houieong. (From left to right: Lead I, Lead II, Lead III)

ecen3003 lead1-wuyufei.adicht: Chart View (PowerLab 26T)				ecen3003 lead2-wuyufei.adicht: Chart View (PowerLab 26T)				ecen3003 lead3-wuyufei.adicht: Chart View (PowerLab 26T)			
ecen3003 lead1-wuyufei.adicht: Comments*				ecen3003 lead2-wuyufei.adicht: Comments*				ecen3003 lead3-wuyufei.adicht: Comments*			
Display channels: 1-8 Filter:				Display channels: 1-8 Filter:				Display channels: 1-8 Filter:			
Channel	Number	Time	Comment	Channel	Number	Time	Comment	Channel	Number	Time	Comment
5	5	3.688 s	P	5	5	4.446 s	P	5	5	8.292 s	P
5	4	3.794 s	Q	5	4	4.573 s	Q	5	4	8.411 s	Q
5	1	3.825 s	R	5	1	4.606 s	R	5	1	8.446 s	R
5	2	3.843 s	S	5	2	4.626 s	S	5	2	8.507 s	S
5	3	4.064 s	T	5	3	4.817 s	T	5	3	8.661 s	T
								2	6	8.67 s	R1
								5	8	9.148 s	R2
								2	7	9.383 s	R2

Fig.10. The information of the labeled QRS Complex of the ECG signal obtained from volunteer 2 Wu Yufei. (From left to right: Lead I, Lead II, Lead III)

As shown in Figures 9 and 10, for volunteer 1, the blood flow trace got from the finger pulse transducer has $X1=9.06s$ and $X2=10.053s$. The ECG signal of “R” point got through the basic method of measuring Einthoven Limb Leads has $R1=4.673s$ and $R2=5.69s$. For volunteer 2, the blood flow trace got from the finger pulse transducer has $X1=8.67s$ and $X2=9.383s$. The ECG signal of “R” point got through the basic method of measuring Einthoven Limb Leads has $R1=8.446s$ and $R2=9.148s$.

Therefore, the heart rate from the recorded ECG signal:

$$\frac{1}{5.69 - 4.673} \times 60 = 59 \text{ rpm (Volunteer 1)}$$

$$\frac{1}{9.148 - 8.446} \times 60 = 86.455 \text{ rpm (Volunteer 2)}$$

The heart rate from the recorded pulse wave:

$$\frac{1}{10.053 - 9.06} \times 60 = 60.423 \text{ rpm (Volunteer 1)}$$

$$\frac{1}{9.383 - 8.67} \times 60 = 84.15 \text{ rpm (Volunteer 2)}$$

6. What does the peak in the blood flow trace represent?

In the blood flow trace recorded by the finger pulse transducer, the peak usually represents the highest point in intra-arterial pressure when the heart contracts, which the ventricles contract and pump blood into the arteries, causing a rapid rise in intra-arterial pressure. The height of the peak may reflect the force with which the heart pumps blood and the elasticity of the arteries. Higher peaks may indicate good heart function, while lower peaks may suggest problems such as weakened heart function or atherosclerosis.

7. Calculate the time differences between peaks in the blood flow trace, are they equal to the corresponding time differences between R waves in ECG traces?

As mentioned in Q4 and 5, from Figures 9 and 10, we found the time differences between peaks in the blood flow traces:

$$\Delta X1=10.053-9.06=0.993 \text{ s (Volunteer 1); } \Delta X2=9.383-8.67=0.713 \text{ s (Volunteer 2),}$$

also the corresponding time differences between R waves in ECG traces:

$$\Delta R1=5.69-4.673=1.017 \text{ s (Volunteer 1); } \Delta R2=9.148-8.446=0.702 \text{ s (Volunteer 2)}$$

After comparison, we can find that the time differences between peaks in the blood flow traces are approximately the same as the corresponding time differences between R waves in ECG traces.

C. Critical thinking

1. Is it possible to measure ECG without following the standard lead locations? What is the consequences?

Yes, it is possible. The ECG signals can still be recorded if the electrodes are not placed at the standard lead locations. However, the recorded signal will differ from the standard ECG traces because the electrical activity of the heart is being captured from different angles, which may lead to Mis being missed and that “silent areas” of the heart’s electrical activity may not be detectable.

In this case, the measurement will lead to the following consequences:

1. Waveforms will be changed from our standard. Since the ECG waveform (P wave, QRS complex, T wave, etc.) depends on the position of the electrodes. If electrodes are placed incorrectly, the recorded waveforms may be distorted or not represent the true electrical activity of the heart, making it difficult to identify abnormalities in the heart.

2. Loss of Diagnostic Value. Standard lead placement ensures that the ECG captures the heart's electrical activity from specific, predefined angles, enabling accurate diagnosis. Non-standard placement may obscure critical diagnostic features, leading to false negatives or false positives.

3. Inconsistent Results. Standard lead positions allow test results to be repeated at different times and between different healthcare facilities. If lead positions deviate from the standard, it is difficult to compare ECG results.

4. Reduced Sensitivity: Certain cardiac abnormalities may not be detected by incorrect lead placement. (this may be due to the “silent areas” that can’t be detected)

III. LABORATORY #2

Electromyogram (EMG) and Muscle fatigue analysis

A. Objectives

To analyze muscle fatigue through the recorded EMG (Electromyogram).

B. Overview of principles and procedures

1. Make sure the volunteers don't have any accessories in their hands.
2. The Bio amp has one connection, CH1(+, -), CH2(+, -) and ground. Connect the dry grounding strap to the ground of the Bio amp and tie it to the volunteer's wrist, with the fluffy part close to the skin. CH1(+, -) is connected to a Disposable ECG recording electrodes, and two Disposable ECG recording electrodes are attached to the volunteer's biceps muscle, 2-5cm apart. CH2(+, -) was attached to the volunteer's triceps in the same manner. Shown as Figure 1.

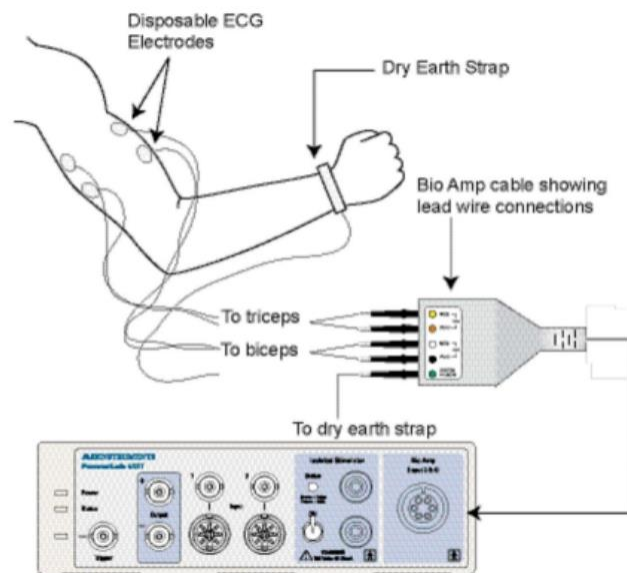


Fig.1. Experimental device connection

3. Monitoring the EMG of the biceps and triceps of the participants through Powerlab, the participants were allowed to continue biceps bending with dumbbells until exhaustion and were analyzed for exhaustion through the spectrum.
4. Fatigue signals in the biceps and triceps of subjects were observed by analyzing the frequency of EMG signals.

C. Results

1. Show all the measurement results during the experiment.

The participant performed biceps bending using 2.5kg dumbbell plates for almost 3 minutes in this experiment. Figures 2 and 3 show the EMG measurements of the biceps and triceps muscles of the participant, respectively.

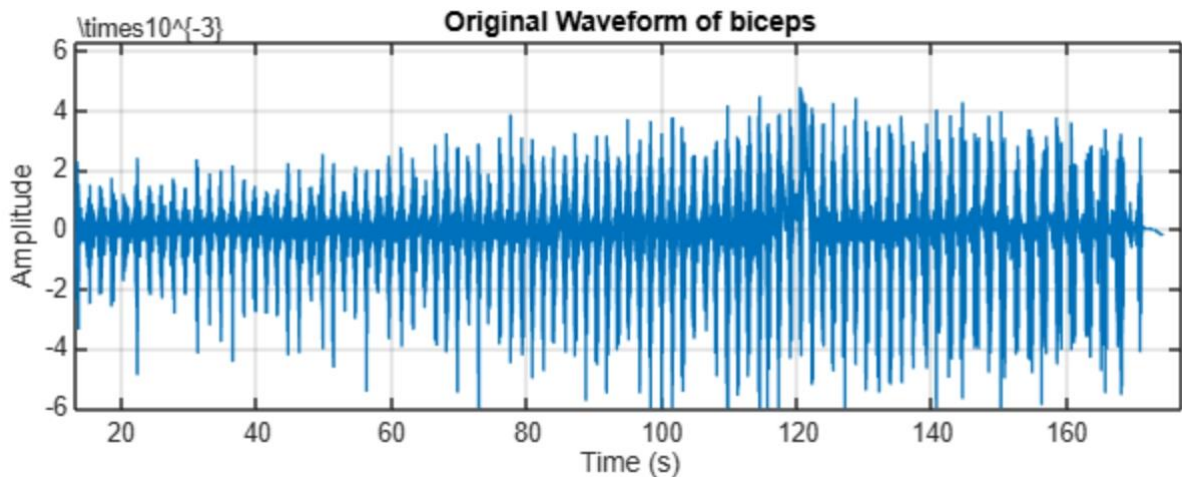


Fig.2. EMG of the biceps muscle

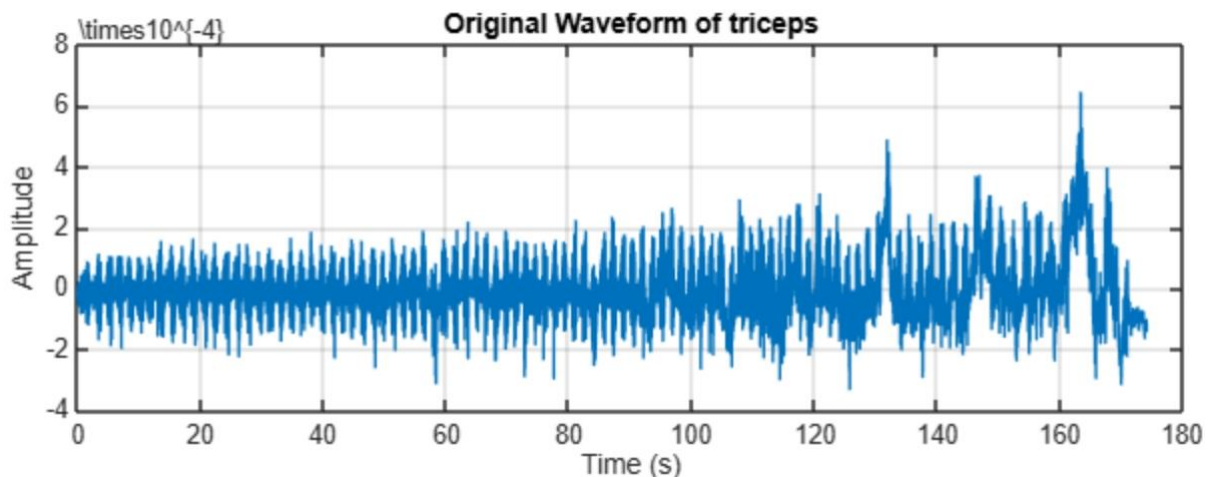


Fig.3. EMG of the triceps muscle

2. Why can't EMG show a regular series of waves like the ECG?

Because the force generated by the muscles is variable, there is a fixed fluctuation in the force export on its own. And the frequency of the dominant oscillation is only 1-2Hz, when we observe the transmission signals of the neural system, we can observe the fluctuation of the force. In addition to this, in muscle control, multiple motor units are driven together. These signals are given out randomly and irregularly, resulting in significant variations in the neuroelectric activity of the muscles under different conditions. This is why EMG signals show more irregular shapes. Whereas, the contraction and diastole

of the heart is cyclic, which is why the ECG waveforms show regularity.

3. By analyzing the median / mean frequencies of EMG, one can identify the progress of fatigue with the change of frequency of EMG. Which one (medium frequency/mean frequency) is better reflect the fatigue?

The spectrum of triceps and biceps muscles were analyzed by Fourier transform (Figure 4&5). We can see that the muscle starts to exert force from 15 seconds, and then the EMG frequency is expressed from 0 to 500 Hz. From the spectrum analysis, we can see that as time goes by, the high frequency component decreases and the low frequency component increases, which is more obvious in the triceps spectrum. This is a sign of muscle fatigue.

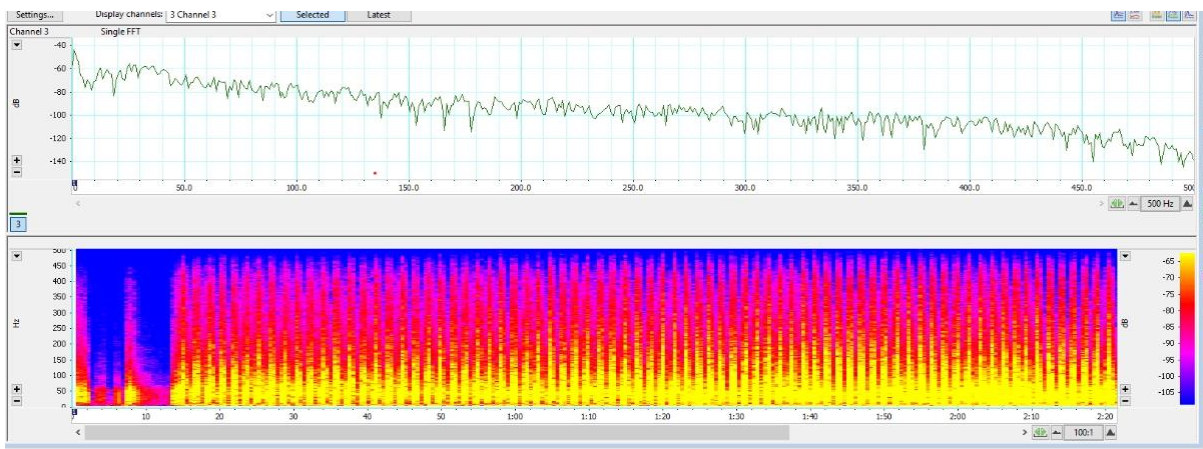


Fig.4. Spectrum analysis of the biceps muscle EMG

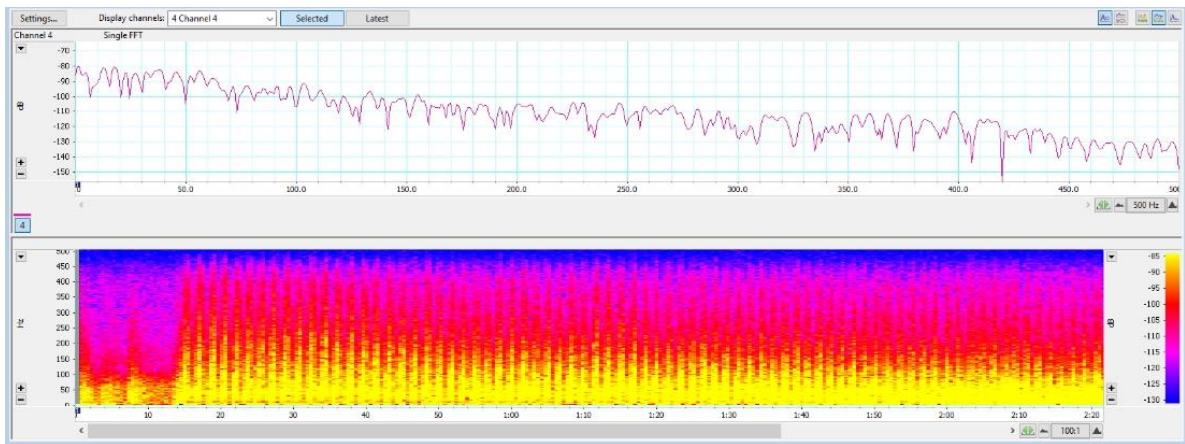


Fig.5. Spectrum analysis of the triceps muscle EMG

To further the performance of muscle fatigue in EMG, we can analyze the mean and median frequency.

$$f_{Mean} = \frac{\sum f * P_1}{\sum P_1}$$

$$f_{Median} = f_k, \text{ where } k \text{ is the smallest index such that } C_k \geq P_{half}$$

$$Biceps \begin{cases} f_{Mean} = 72.6143Hz \\ f_{Median} = 46.5577Hz \end{cases}$$

$$Triceps \begin{cases} f_{Mean} = 63.2624Hz \\ f_{Median} = 37.5758Hz \end{cases}$$

We can also plot the average and median frequencies per second through the software (Figure 6&7). We can see that although there is some fluctuation in the mean frequency and median values, the overall trend is decreasing over time. This is a symptom of muscle fatigue. The average frequency has a more pronounced and stable decreasing trend than the median frequency, so the average frequency is better able to characterize muscle fatigue. Also, the fatigue of the triceps muscle is more obvious.

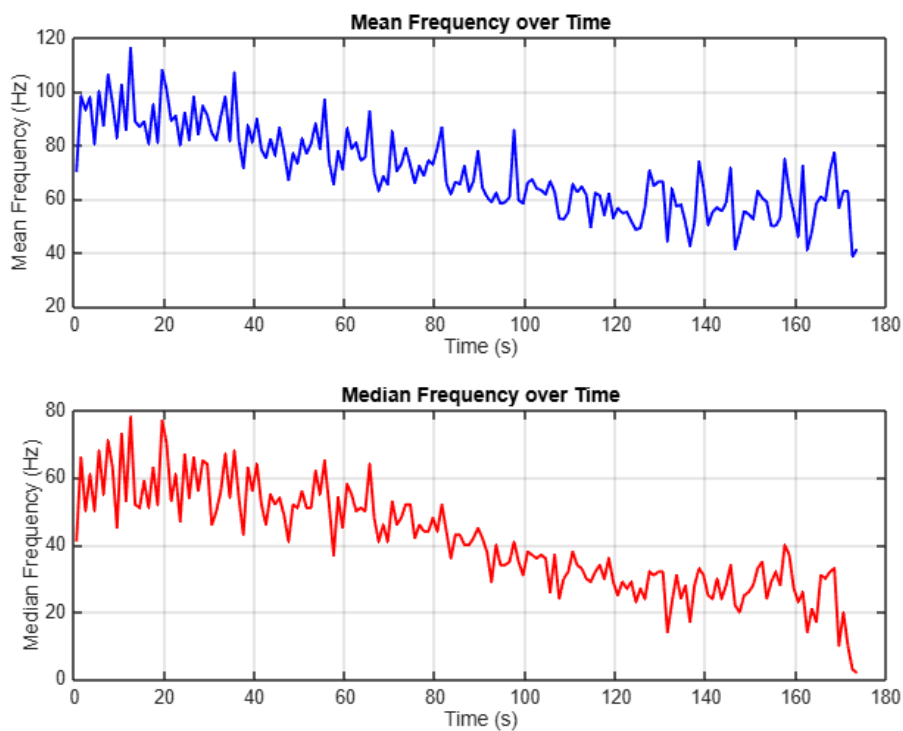


Fig.6. Mean and median frequency plots of the biceps muscle

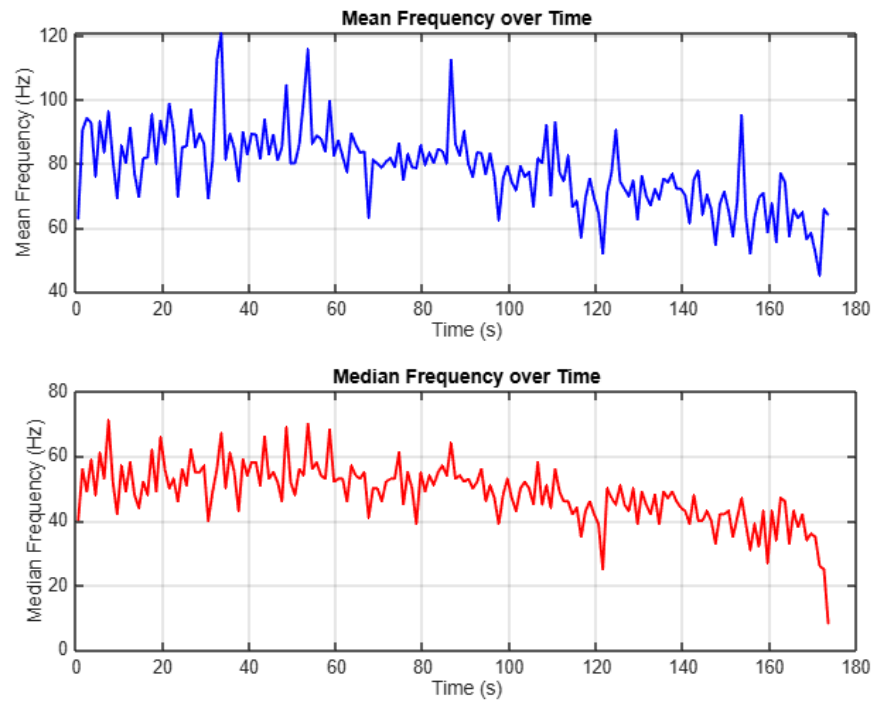


Fig.7. Mean and median frequency plots of the triceps muscle.

D. Critical thinking

1. During the experiment, we can observe EMG activities on both Bicep & Tricep on some people and only EMG activities on Bicep for some other. What are the reason for that?

During the experiment, we could observe the EMG signal of biceps and triceps in the participants, but some experimental groups did not have EMG signal of triceps. In fact, the EMG signal of the triceps in the biceps bend is due to two factors, one is because the participant drops the forearm below the horizontal plane during the centrifugal process of the biceps bend, which induces the triceps to exert force. The second factor is that the subject utilized shoulder force during the contraction of the bicep bend, which resulted in the involvement of the triceps muscle group in the movement, which was incorrect.

In our experiment, the participation of the triceps in the activity belonged to the first factor, and it can be seen that in the original waveform, the timing of the biceps and triceps force does not overlap (Figure 8). This also explains why the fatigue tendency of the triceps is more pronounced than that of the biceps, since the weight of the dumbbells is not particularly heavy for the biceps to exert force, but rather the triceps are more heavily loaded during the centrifugal process.

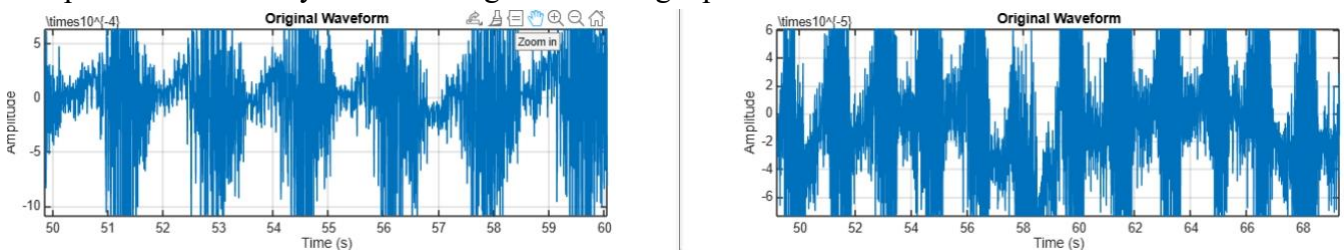


Fig.8. EMG signal in the biceps and triceps at 50 seconds to 60 seconds

IV. LABORATORY #3

Blood pressure measurement

A. Objectives

Familiar with Oscillometric and Palpatory methods and the measurement of blood pressure.

B. Overview of principles and procedures

This experiment requires us to measure the blood pressure (also determine the systolic and diastolic pressure) through both Oscillometric and palpatory methods. Before the measurement, since the output of the instruments is measured in millivolts, we need to do the conversion first as shown in Figure 1 between the unit of the pressure “mmHg” and the unit “millivolts” by the linear relationship between them, which we got from a prelab we did on a cylindrical paper tube. After the instruments are connected, we apply a high pressure of around 200 mmHg and then slowly deflate the cuff at a rate of 1 to 2 mmHg per second until the cuff deflated. Record the pressure signals measured by the cuff and the blood flow trace measured by the finger pulse transducer.

For the Oscillometric method, a high-pass filter should be applied to the pressure signal. Then we can find the oscillometric part of the signal during the deflation of the cuff’s pressure. As we learned in class, We can find the maximum oscillation O_m during the oscillating part. The location of Systolic oscillations O_s and location of Diastolic oscillation satisfy the relationships as follows:

$$\frac{O_s}{O_m} = 0.55; \frac{O_d}{O_m} = 0.85$$

Therefore, the pressures corresponding to the cuff at the time O_s and O_m occur are the systolic blood pressure and the diastolic blood pressure accordingly.

For the palpatory method, the pulse signal will disappear when the cuff’s pressure is larger than the systolic blood pressure. Therefore, we can determine the systolic blood pressure at the beginning of the first pulse signal during the deflation of the cuff’s pressure. However, this kind of method can’t detect the diastolic blood pressure.

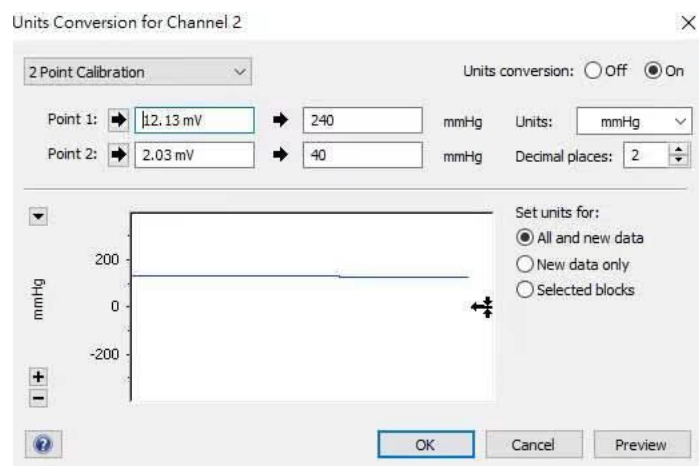


Fig.1. The conversion relationship between the units.

C. Results

1. Show all the measurement results during the experiment.

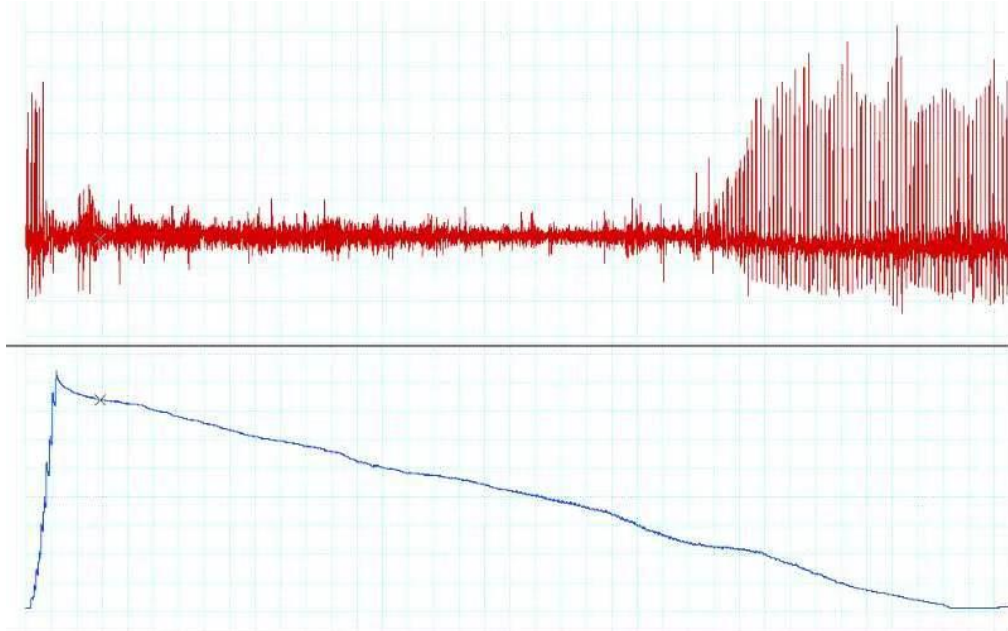


Fig.2. The pulse signal measured by the finger pulse transducer (upper part) and the pressure signal measured from the cuff (lower part).

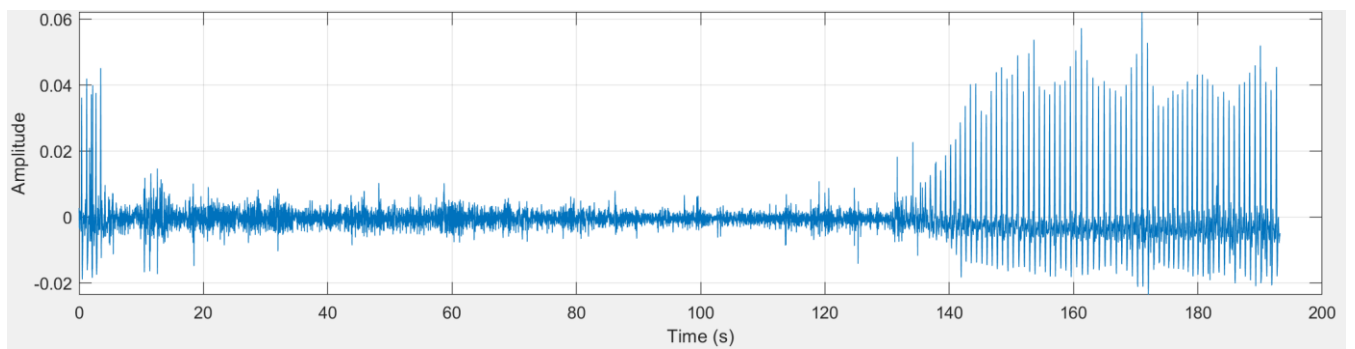


Fig.3. The pulse signal measured by the finger pulse transducer shown in MATLAB.

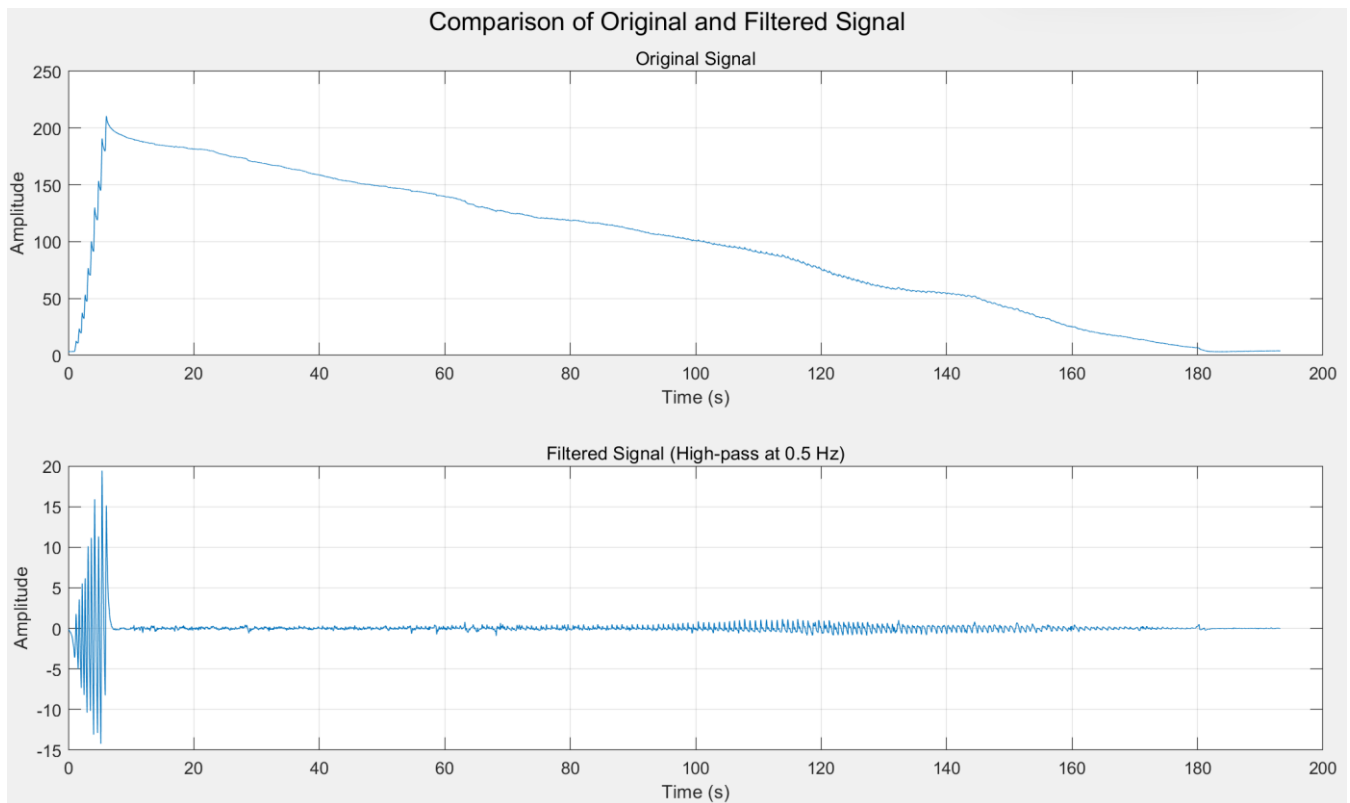


Fig.4. The Comparison of the original and filtered pressured signal measured from the cuff in MATLAB.

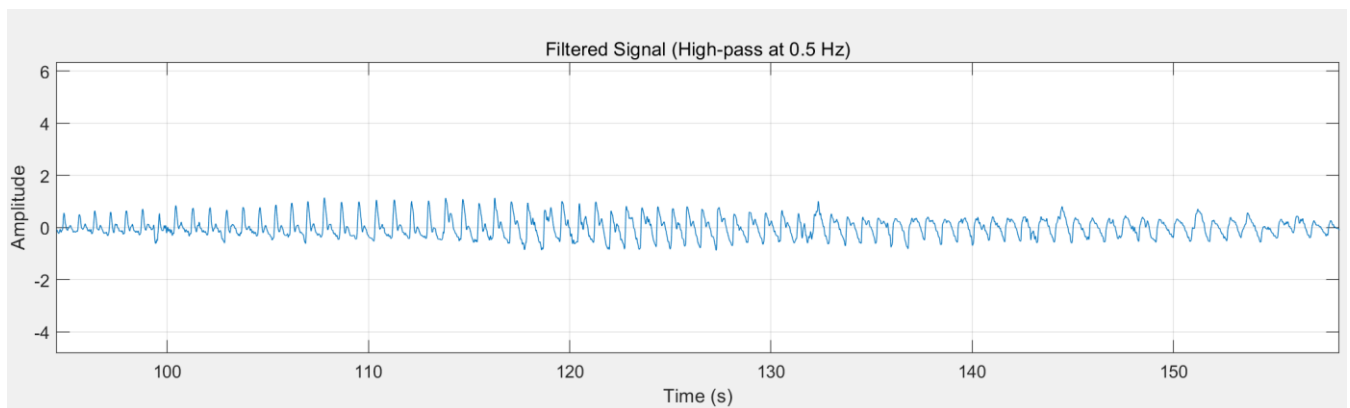


Fig.5. A more evident version of the oscillating part in MATLAB.



Fig.6. The blood pressure measurement result with automatic blood pressure meter.

2. Please calculate the systolic blood pressure. (Om: Maximum oscillations Os: Location of systolic oscillations Od: Location of diastolic oscillations)

For the Oscillometric method, we found the Om, and then determined Os and Od in the oscillating part as shown in Figure 7 according to their relationship mentioned before, which satisfy:

$$Om = 0.94892 - (-0.868714) = 1.817634$$

$$Os = 1.817634 * 0.55 = 0.9996987 \approx 0.733656 - (-0.272285) = 1.005941$$

$$Od = 1.817634 * 0.85 = 1.5449889 \approx 1.00792 - (-0.548377) = 1.555667$$

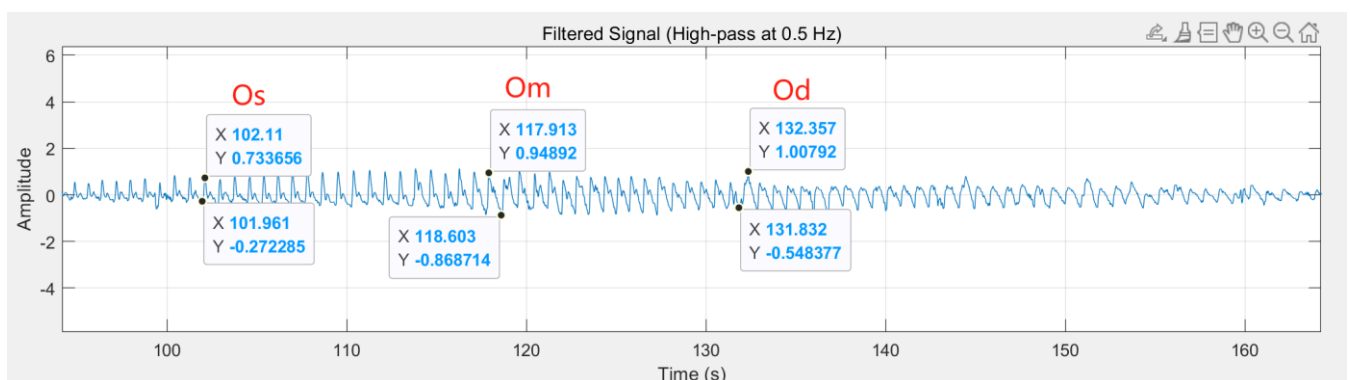


Fig.7. Om, Os, Od determined in the oscillating part.

Then we can find the systolic blood pressure = 99.8906 mmHg and the diastolic blood pressure = 59.3317 mmHg as shown in Figure 8 below:

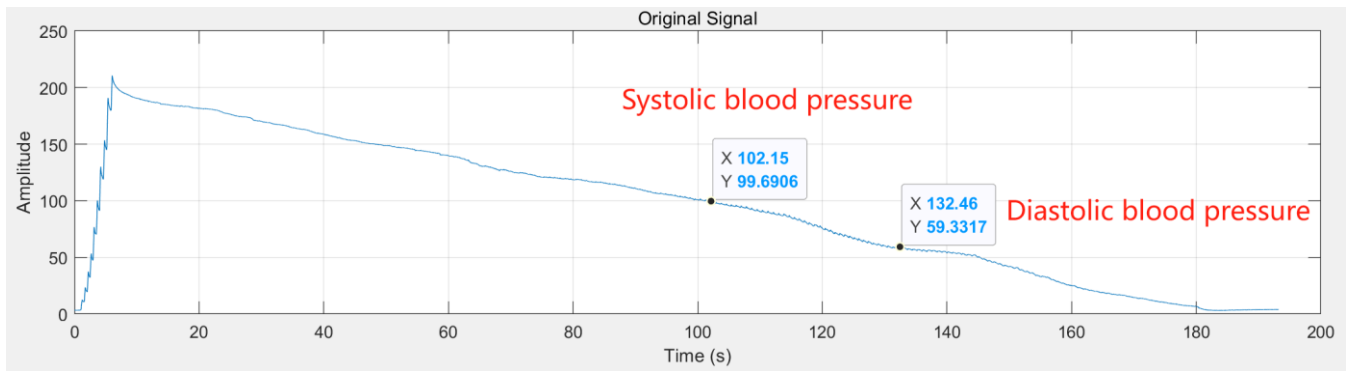


Fig.8. The Systolic blood pressure and Diastolic blood pressure determined by Oscillometric method.

For the palpation method, we can also find the Systolic blood pressure = 58.7005 mmHg as shown in Figure 9.

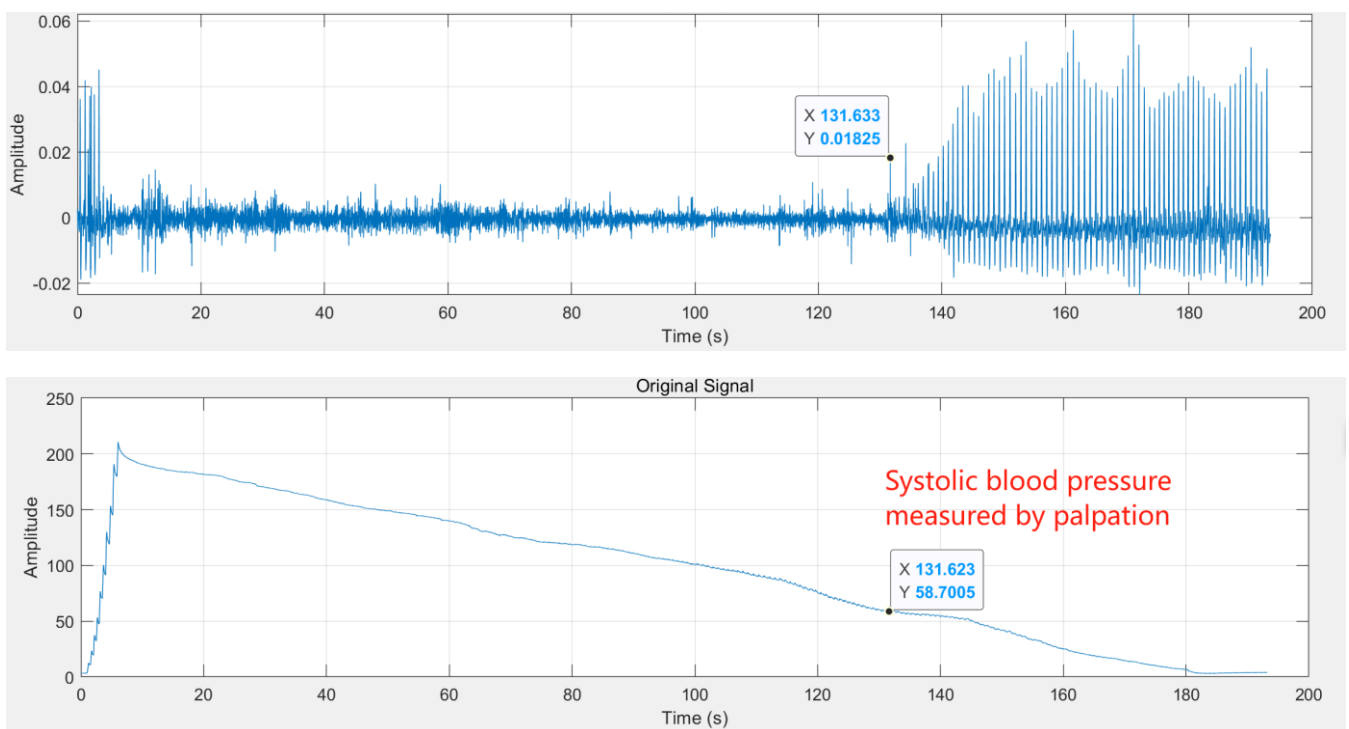


Fig.9. The Systolic blood pressure and Diastolic blood pressure determined by palpation method.

3. *Please compare your measurement with measurement from automatic blood pressure meter.*

From Figure 6, the Systolic blood pressure and Diastolic blood pressure measured by the automatic blood pressure meter are 96mmHg and 58 mmHg respectively. Which are similar to our results of two measurement methods above. Errors in experiments can arise from various factors, some of which are beyond control.

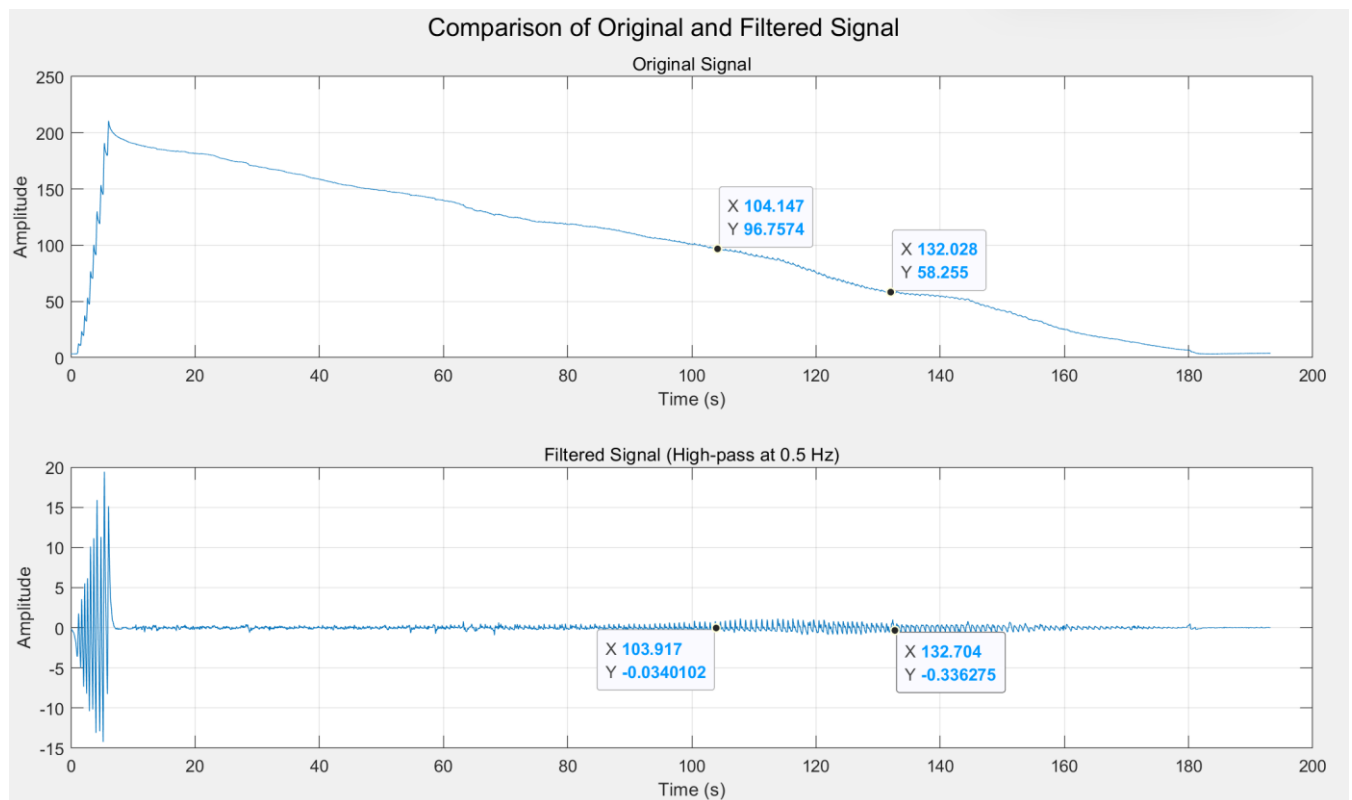


Fig.10. The location of DBP and SBP in oscillating part is we use the data measured by the auyomatic blood pressure meter.

D. Critical thinking

1. Why diastolic blood pressure is not measured by palpation method?

Diastolic blood pressure is not measured by the palpation method primarily due to the limitations of the palpation technique. As we mentioned in “Part B”, the pulse signal will disappear when the cuff’s pressure is larger than the systolic blood pressure. Therefore, we can determine the systolic blood pressure at the beginning of the first pulse signal during the deflation of the cuff’s pressure. However, this kind of method can’t detect the diastolic blood pressure.

V. LABORATORY #4

Electromyogram (EMG) and Muscle loading analysis

A. Objectives

To analysis the EMG-loading relationship of the biceps muscle with different joint angle.

B. Overview of principles and procedures

1. Make sure the volunteers don't have any accessories in their hands.
2. The Bio amp has one connection, CH1(+, -), CH2(+, -) and ground. Connect the dry grounding strap to the ground of the Bio amp and tie it to the volunteer's wrist, with the fluffy part close to the skin. CH1(+, -) is connected to a Disposable ECG recording electrodes, and two Disposable ECG recording electrodes are attached to the volunteer's biceps muscle, 2-5cm apart. CH2(+, -) was attached to the volunteer's triceps in the same manner. Shown as Figure.1

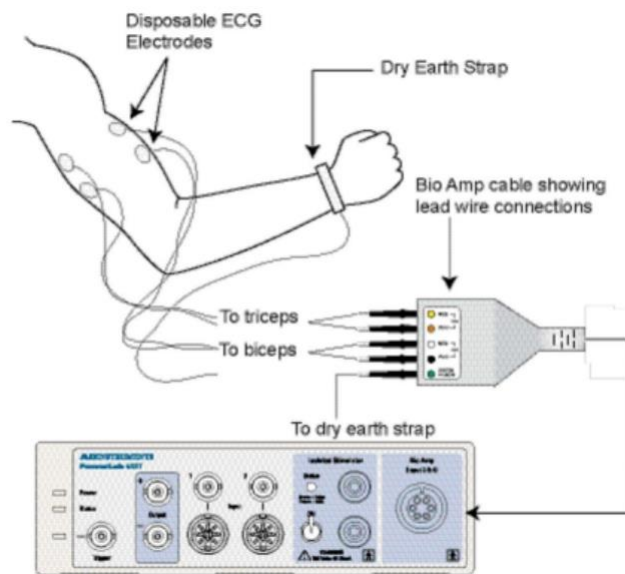


Fig.1. Experimental device connection

3. Fix the joint angle of arm at 180° , 135° , 90° , and 45° respectively (Figure 2). And load the subject with increasing loading (dumbbells or equivalent) from 0kg to 80% of the maximum load the subject can bear. Record the EMG of the biceps for each session. The EMG measurement for each loading should be at least 5s. Note that muscle fatigue will affect the EMG of the biceps muscle, therefore, leave enough rest time for each session of measurement.

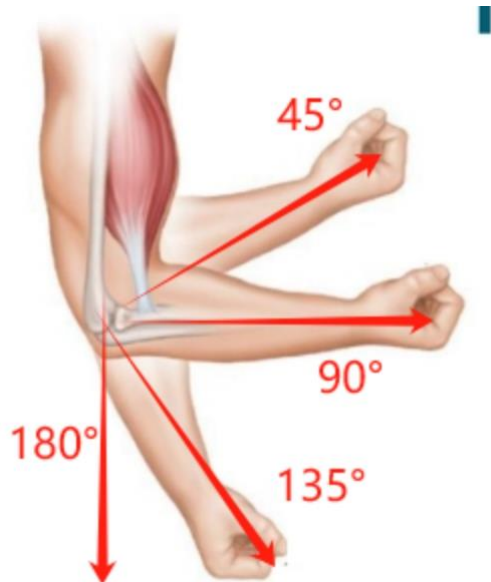


Fig.2. Different angle of arm

4. Extract the EMG values for different loads from the EMG result and calculate the RMS value of the loaded EMG by RMS

C. Results

1. Show all the measurement results during the experiment.

Because of the lack of equipment, we used a pattern of different weights of books in a backpack instead of dumbbells. There were a total of 5 weight processes per measurement (one with the backpack and 4 with the books). Figure 3 represents biceps EMG signal at different angles, while figure 4 represents triceps EMG signal at different angles.

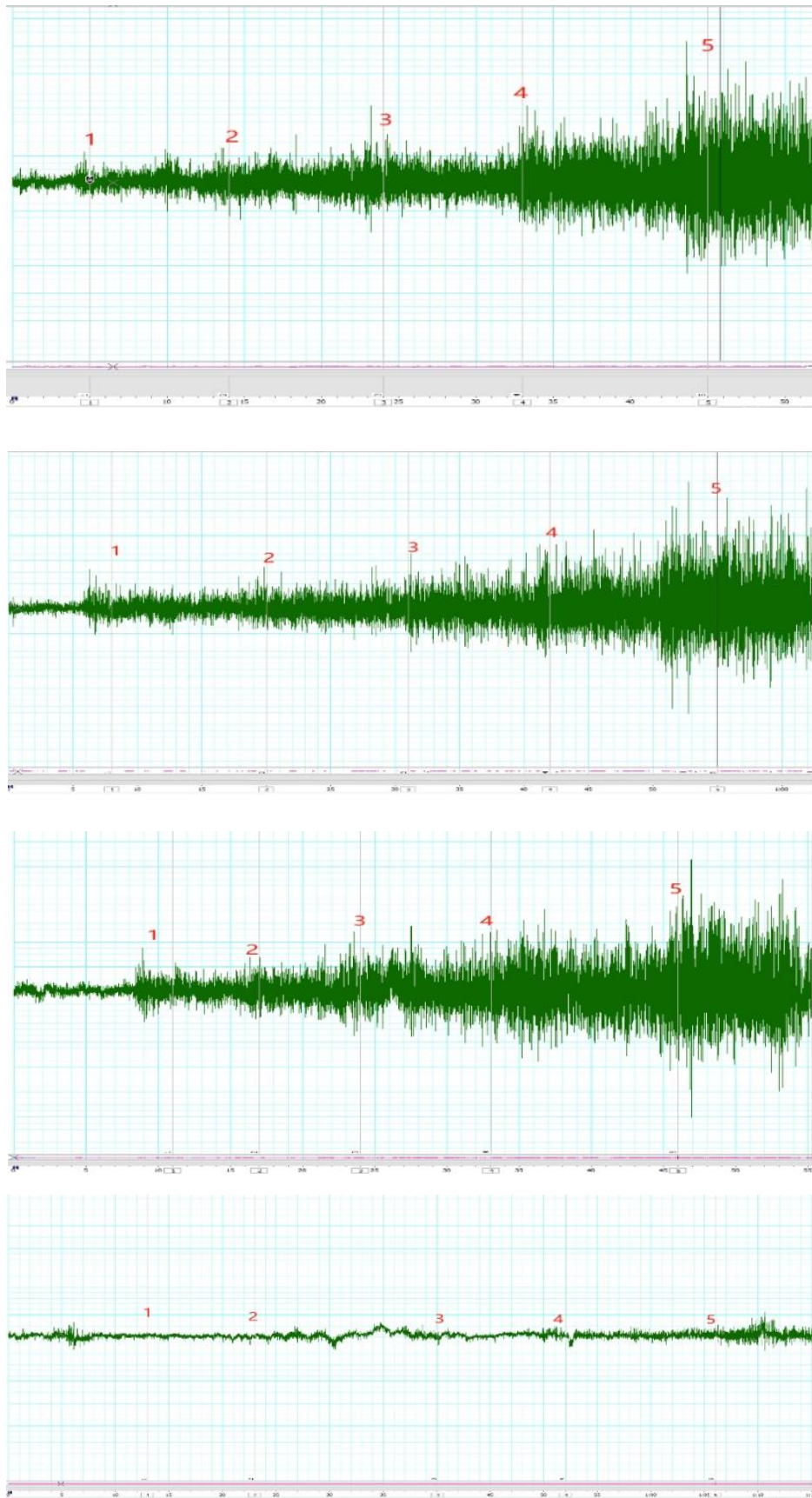


Fig.3. EMG signal of Biceps (Top to Bottom: 45°, 90°, 135°, 180°)



Fig.4. EMG signal of Triceps (Top to Bottom: 45°, 90°, 135°, 180°)

2. What is the relationship of the muscle loading with the amplitude of EMG?

We can analyze the relationship between loading and muscle performance by analyzing the RMS values of EMG signals during different loading periods.

$$RMS = \sqrt{\frac{1}{n}(x_1^2 + x_2^2 + \dots + x_n^2)}$$

Figure 5-8 shows load-RMS plots of EMG signals of biceps at different angles, and Figure 9-12 shows load-RMS plots of EMG signals of biceps at different angles. Since we used books instead of dumbbells as the load, we were unable to quantify the weight. Despite this, we can still see that in biceps EMG performance, the EMG amplitude still rises with load. While we can't directly see if the load and EMG are linearly related, we can still see through a couple of graphs that have about the same ratio of distance between EMG and the linear trend line. So from the EMG analysis of the biceps, the RMS of the EMG was linearly correlated with the loaded weight of the participant. However, the EMG analysis of the triceps shows essentially no linear correlation, perhaps because the triceps are less involved in force generation.

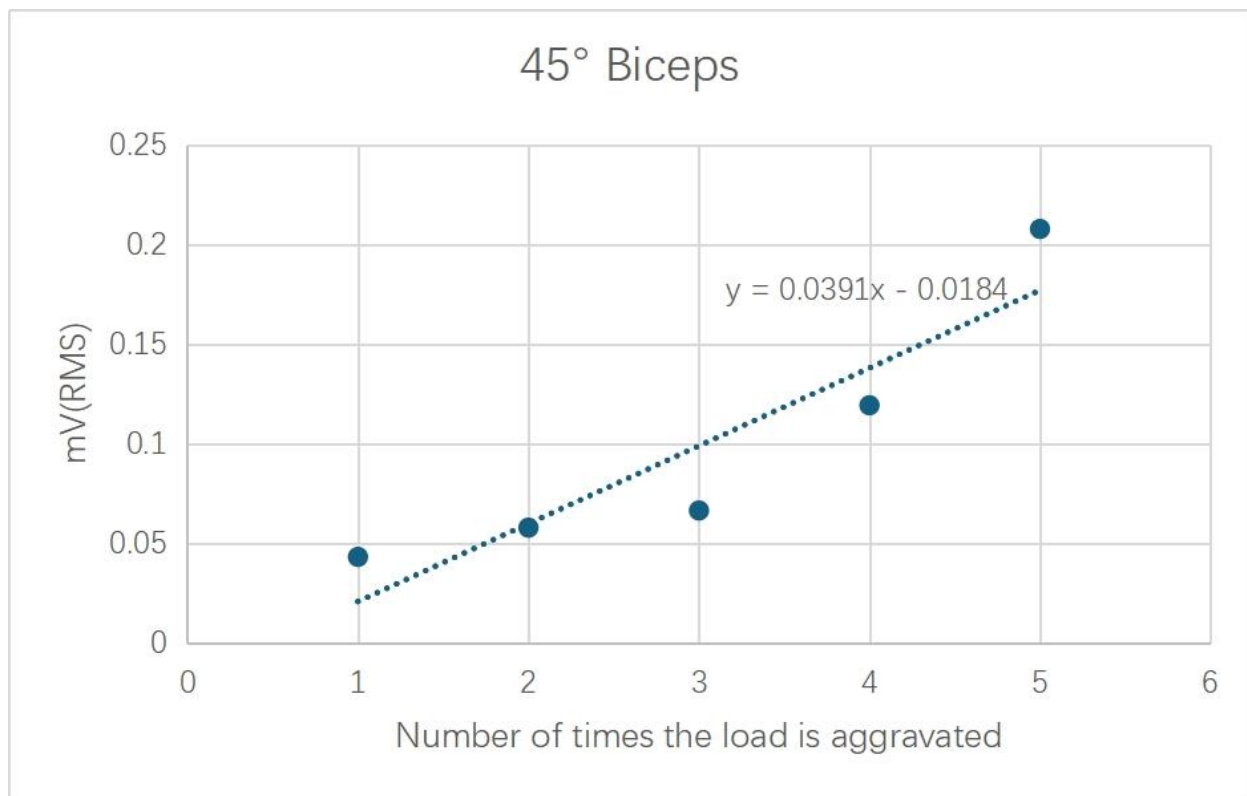


Fig.5. Biceps EMG signal at 45°. (RMS values at different loading intervals)

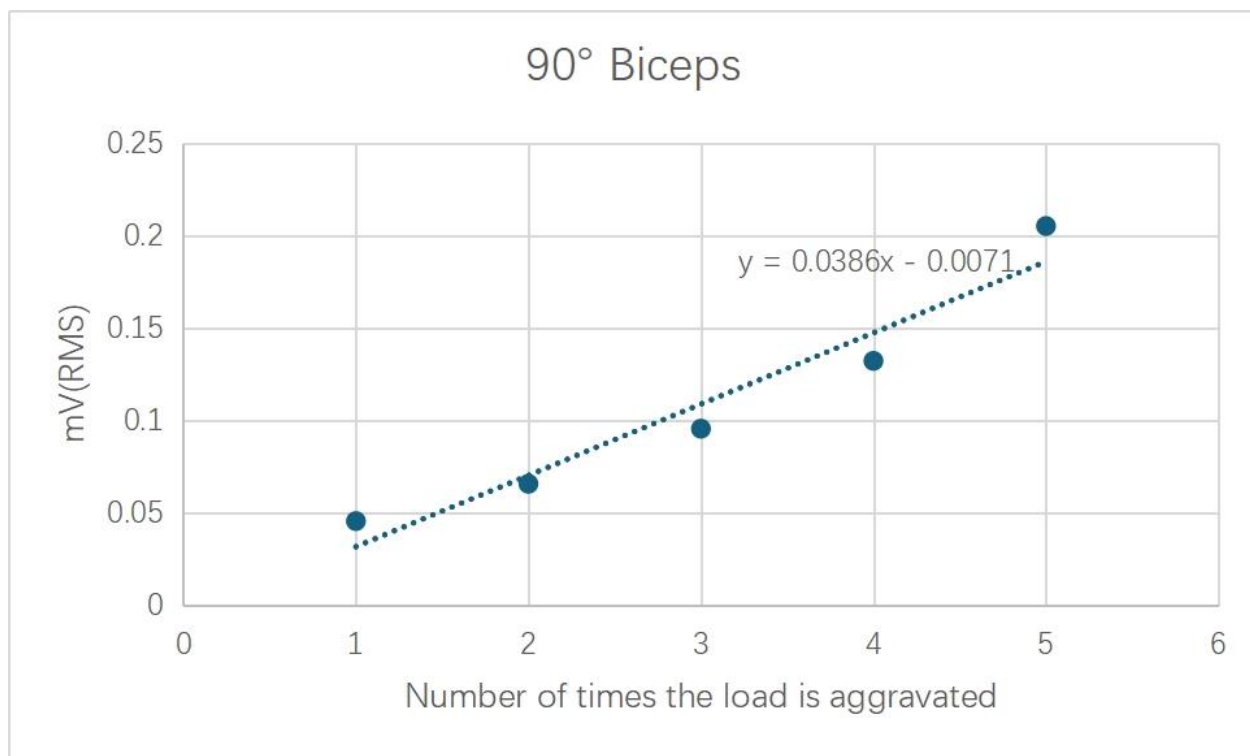


Fig.6. Biceps EMG signal at 90°. (RMS values at different loading intervals)

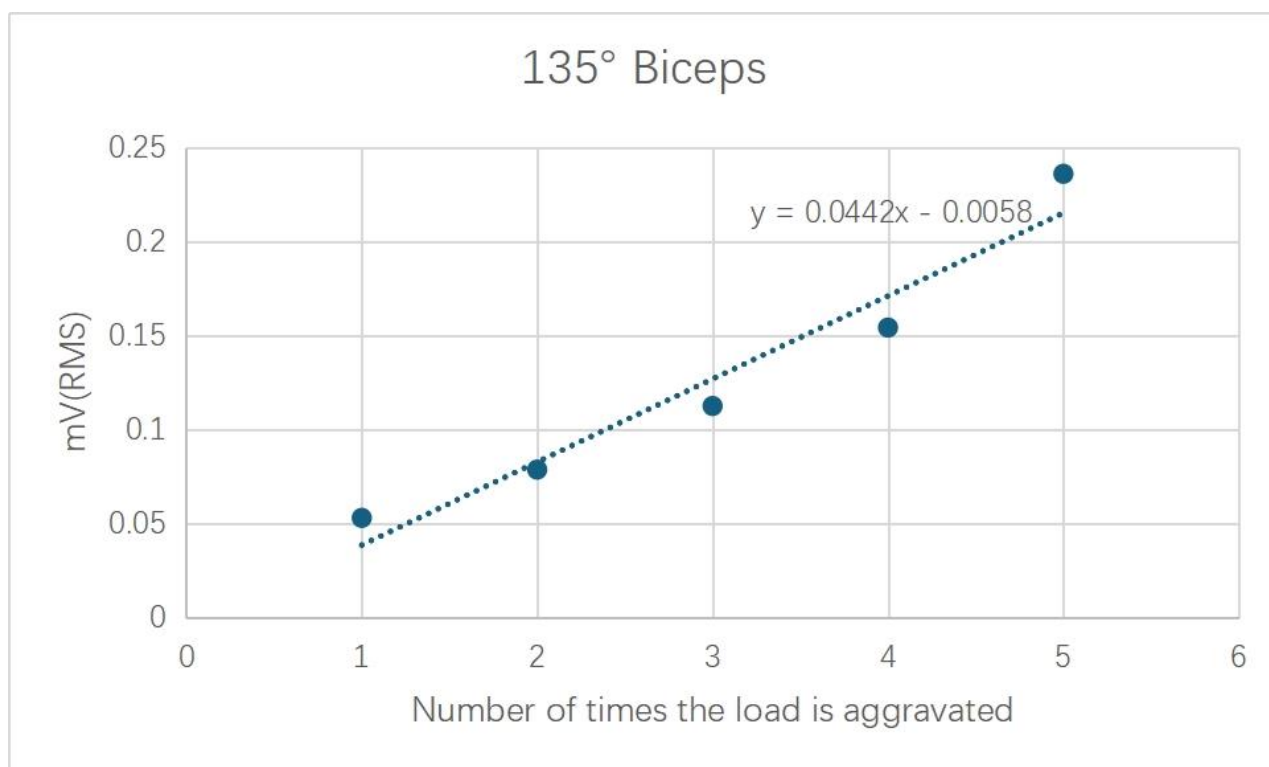


Fig.7. Biceps EMG signal at 135°. (RMS values at different loading intervals)

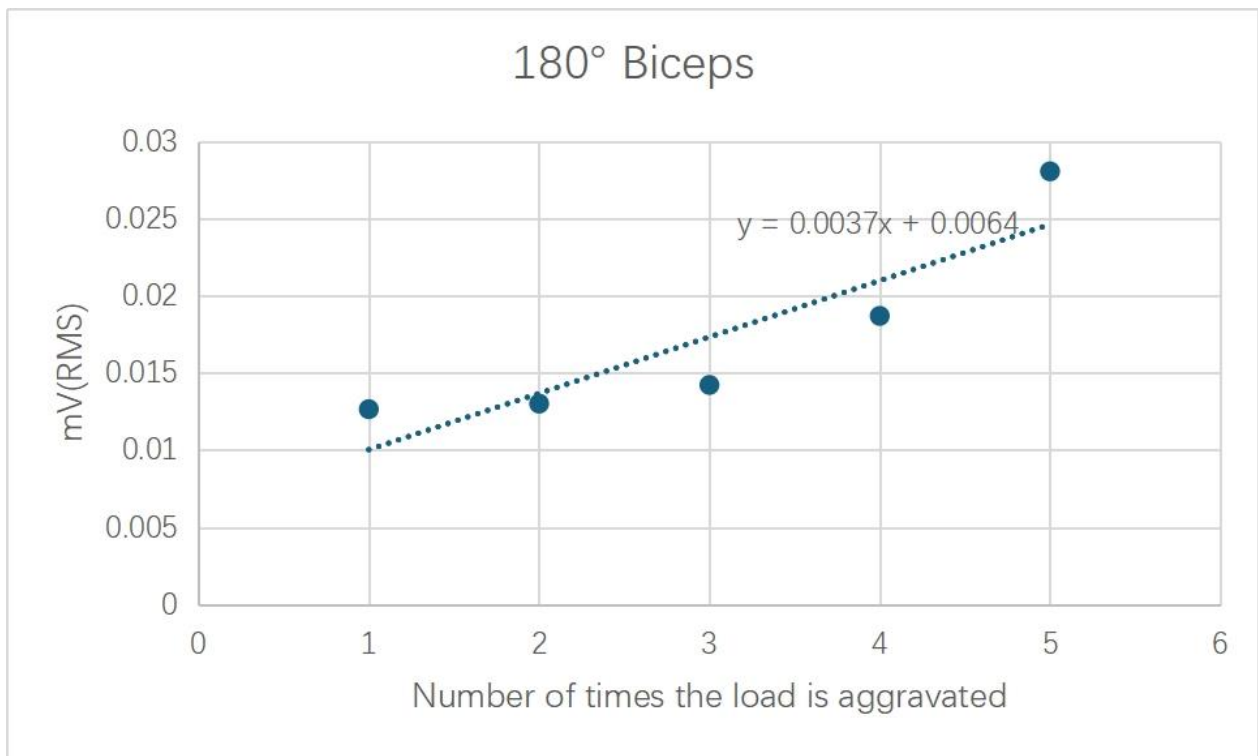


Fig.8. Biceps EMG signal at 180°. (RMS values at different loading intervals)

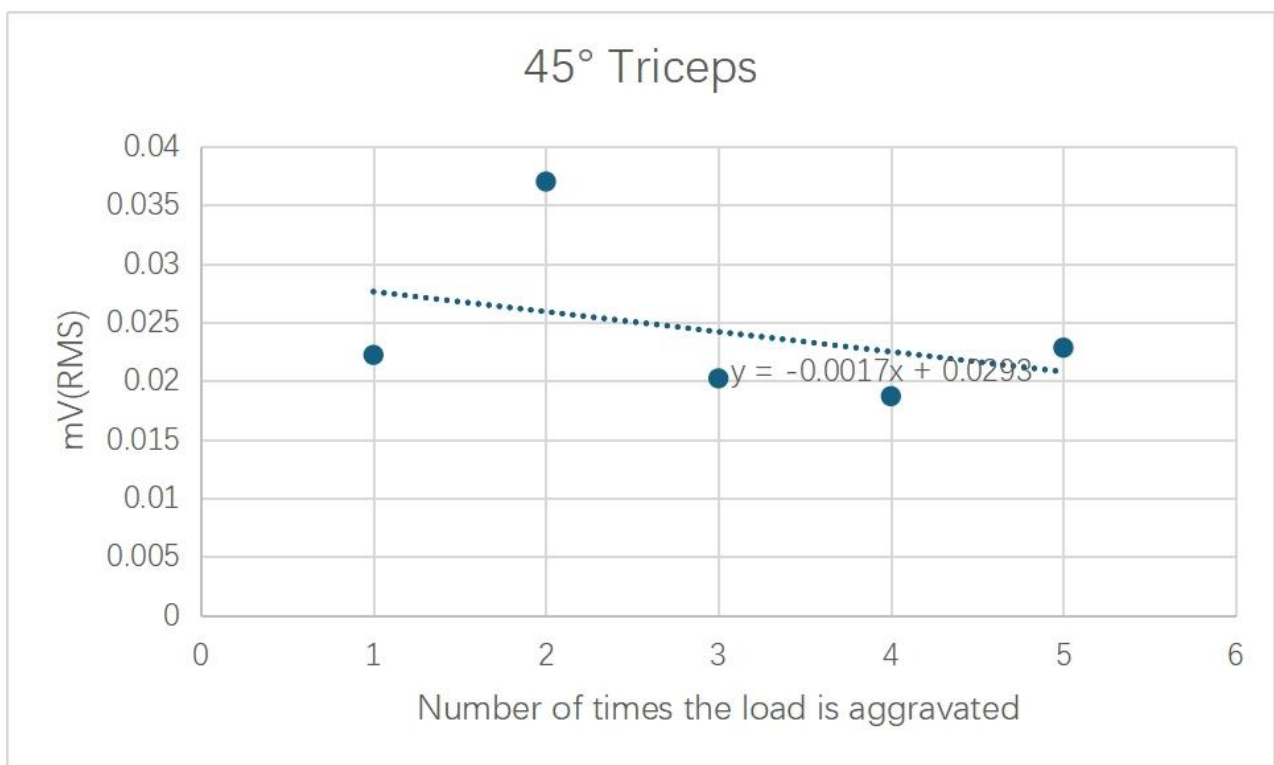


Fig.9. Triceps EMG signal at 45°. (RMS values at different loading intervals)

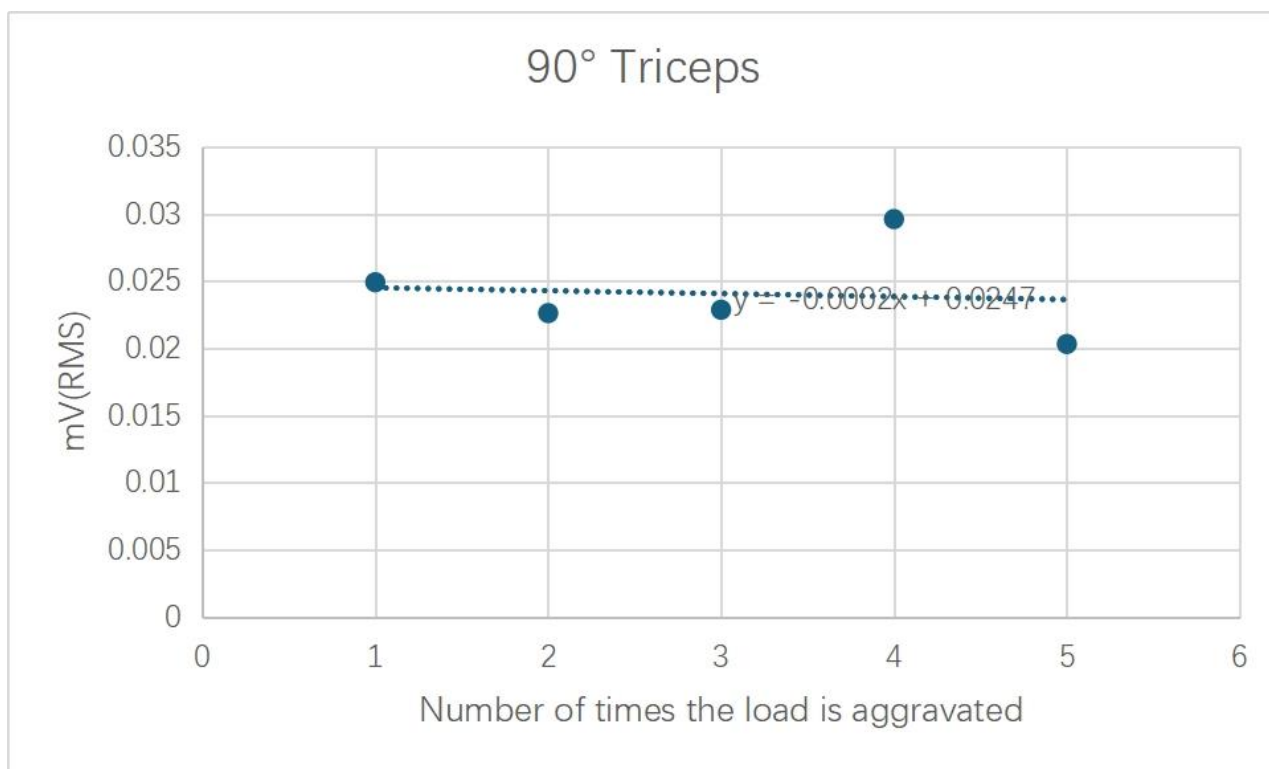


Fig.10. Triceps EMG signal at 90°. (RMS values at different loading intervals)

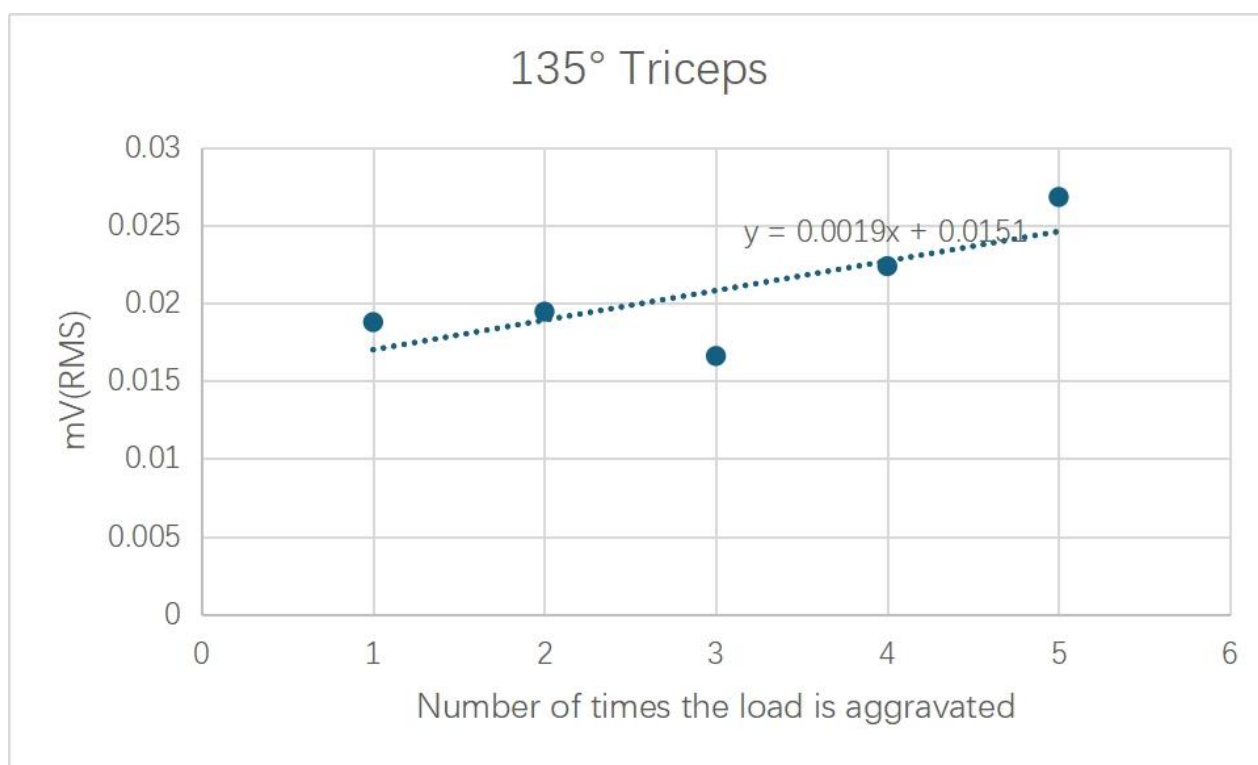


Fig.11. Triceps EMG signal at 135°. (RMS values at different loading intervals)

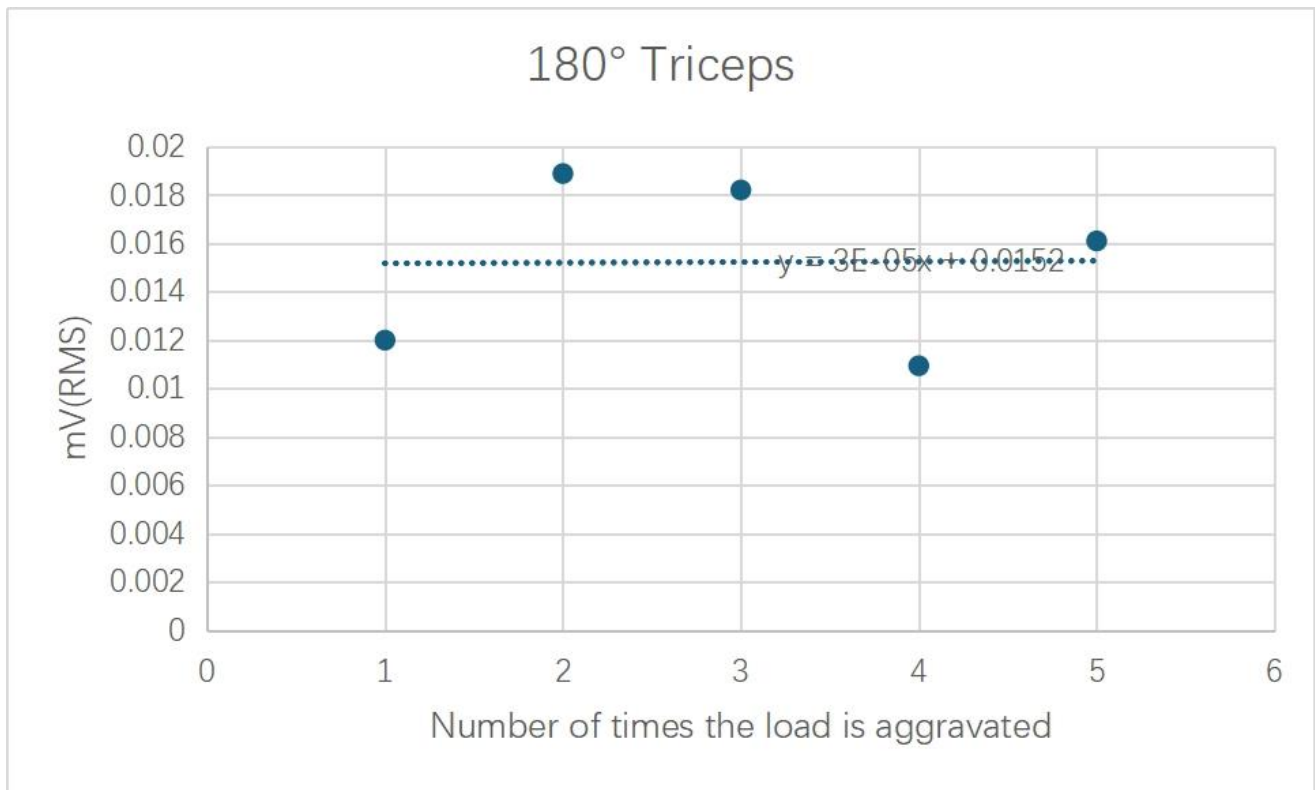


Fig.12. Triceps EMG signal at 180°. (RMS values at different loading intervals)

3. *Can the joint angle affect the force generated by the Biceps muscle? Why?*

Review the EMG signals of biceps and triceps at different joint angles (Figure 3&4). It is easy to see that the joint angle affects the performance of the biceps and triceps. Especially when analyzing the EMG signals of the biceps, it is noticed that the EMG amplitude is higher at 45° and 135°, while the EMG amplitude is very low at 180°. From a physical point of view, our arm is a moment of force, so it takes varying amounts of force to lift objects from different angles. In this experiment, the required force was mainly provided by the biceps muscle in 45°, 90° and 135° joint angles. In 180° joint angles, the required force was mainly provided by the forearm flexor muscle group.

$$\tau = F * r * \sin\theta$$

In this experiment, the required force was mainly provided by the biceps muscle in 45°, 90° and 135° joint angles. In 180° joint angles, the required force was mainly provided by the forearm flexor muscle group.

And from a biological point of view, different joint angles will make the muscles have different levels of stretch. Different lengths of muscle produce different levels of tension, but this is not a linear relationship, and overstretching of the muscle can also result in a loss of muscle strength.

D. *Critical thinking*

1. *Can the joint angle affect the force generated by the Biceps muscle? Why?*

Muscle fatigue can seriously affect the experimental results of this experiment. We are trying to observe the

relationship between RMS of EMG and load weight. Normally the relationship between RMS of EMG and load weight is positive because more force needs to be provided. And when muscle fatigue occurs, the strength of the muscle decreases. Subjects will experience postural changes that disperse the force so that the EMG amplitude results will change later on in the experiment. It defeats the purpose of the experiment and reduces the research value of the results.

VI. LABORATORY #5

Construction of an ECG amplifier

A. Objectives

The purpose of this experiment is familiar with Electrocardiography (ECG) and Bio potential amplifier. In this lab, students are required to simulate and build an instrumentation amplifier with three op-amps configurations.

B. Overview of principles and procedures

During this Lab, we conduct experiments with different circuits of this instrumentation amplifier with three op-amps configurations to measure Gain_{cm}, Gain_{dm}, and CMRR along with different frequencies, also consider the relationship between resistors and gains.

Due to some irresistible errors in the experiment, the experimental results and the ideal value maybe have some errors. Moreover, the source generator and oscilloscope we used are quite old, and some functions and values may have big errors. For instance, since our generator can not display the output voltage value and the oscilloscope is not accurate when displaying smaller output value (mv), also we can't use large voltage source signal which will lead to clipping. Therefore, we firstly set the voltage source value as 1 V with 30 dB of the gain added so that the 1 V is accurately and clearly showed on our oscilloscope (Figure a in appendix). Then we can calculate the smaller signal without 30 dB gain added that is truly used as our input voltage source. This smaller signal value can be calculated as

$$30 = 20 \log_{10} \left(\frac{1}{V_{small}} \right)$$
$$V_{small} = 31.62 \text{ mV}$$

After that, this generator of the input voltage will remain the same to conduct our experiment to measure the differential mode gain part.

C. Pre-Laboratory work

1. In according to the instrumentation amplifier given in Figure 4, please calculate the differential gain of the amplifier.

According to the formula given in the background information part, the differential gain:

$$\text{Gain } dm = \left(1 + 2 \frac{R4}{R1} \right) \frac{R8}{R6} = \left(1 + 2 \times \frac{330k}{15k} \right) \times \frac{1.2M}{68k} = 794.1176$$

2. Is it possible to calculate the common mode gain of the amplifier with $R_9=200\text{k}\Omega$? If yes, please show your steps. If no, please explain briefly why you cannot.

Sure, when the $R_9=200\text{k}\Omega$, the resistors of the instrumentation amplifier are totally balanced with $R_2=R_3$, $R_4=R_4$, $R_6=R_7$, and $R_8=(R_{10}+R_9)$. In this case, when the input $V_a=V_b$, the output will be 0. Therefore, the common mode gain is 0.

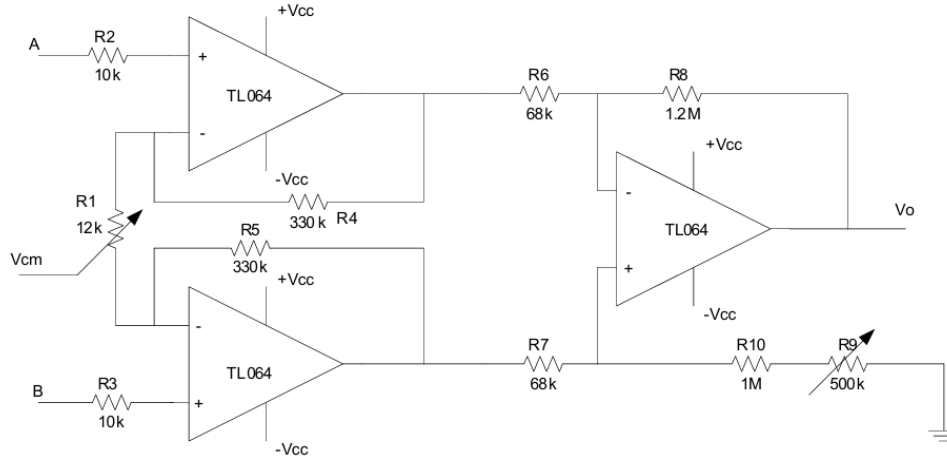


Fig.1. The structure of instrumentation amplifier with three op-amps configurations.

3. Please state the supply voltage (+vcc and -vcc) for the experiment.

The supply voltage throughout our experiment are $\pm 15\text{ V}$.

D. Results

1. Show all the measurement results during the experiment.

a) Differential mode analysis.

After connect the instrumentation amplifier on a prototype board, we can calculate the differential gain in various frequency (100-100k Hz) by measure the input and output voltages. (Table 1) Also we plot the figure of the relationship between the frequency and differential mode gain shown as Figure 2.

f(kHz)	0.05	0.1	0.5	1	5	10	50	100
V _{ipp} (V)	0.031623	0.031623	0.031623	0.031623	0.031623	0.031623	0.031623	0.031623
V _{opp} (V)	23.8	24.4	24.1	23.8	21.9	18.4	3.69	0.97
Ad _m	752.6221	771.5957	762.1089	752.6221	692.5388	581.8591	116.688	30.67409

Table 1. Data of Instrumentation amplifier at frequency(100-100kHz)

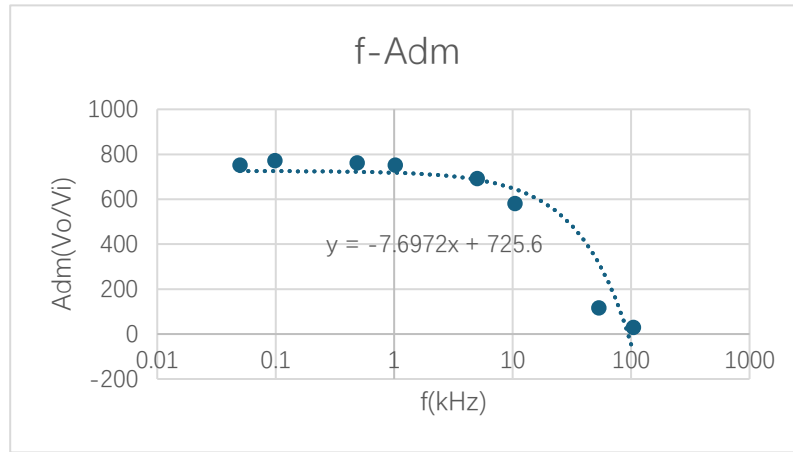


Fig.2. Relationship between the frequency and differential mode gain.

b) Common mode analysis.

After changing the circuit to measure the common mode gain of the amplifier, we found that the frequency will not affect the value of common mode gain as shown below in Figure 3 in part f), we take the result measured at 36k Hz as an example.

$$A_{cm} = \frac{V_{out}}{V_{in}} = \frac{5.75}{12.7} = 0.45$$

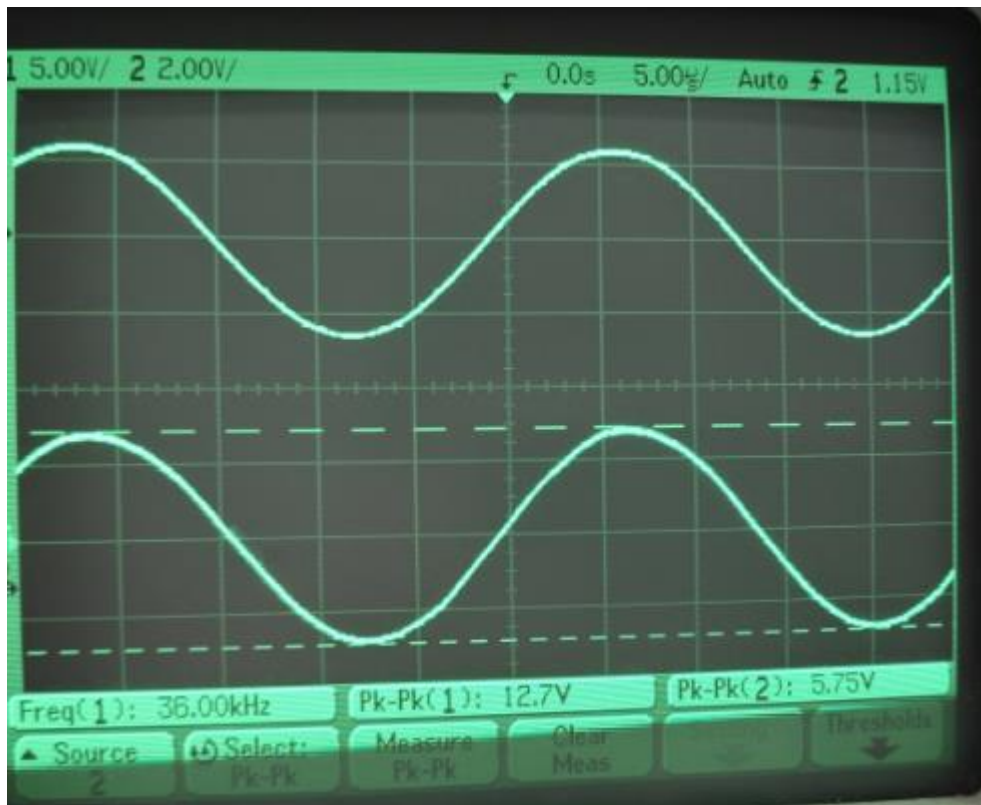


Fig.3. Common mode waveform (f=36kHz, R9 = 500kΩ)

c) Improvement of CMRR

Since the instruments are operating at 50 Hz in Macao, we focus on depressing the noise signal at 50 Hz. By adjusting R9 to different value, we can observe the changing of common mode gain in oscilloscope

and find the minimum value of common mode gain when R9=120 k Ω .

After that, measure the common mode gain (Table 2) and plot the figure of the relationship between the frequency and common mode gain shown as Figure 4, where we can find the common mode gain is around 0.01 V/V.

f(kHz)	0.01	0.05	0.1	1	5	100
V _{ipp} (V)	21.2	21.2	21.2	21.2	21.2	21.2
V _{opp} (V)	0.275	0.284	0.3	0.656	2.59	5.06
A _{cm}	0.012972	0.013396	0.014151	0.030943	0.12217	0.238679

Table 2. Data of Instrumentation amplifier at frequency(100-100kHz), R9 = 120k Ω

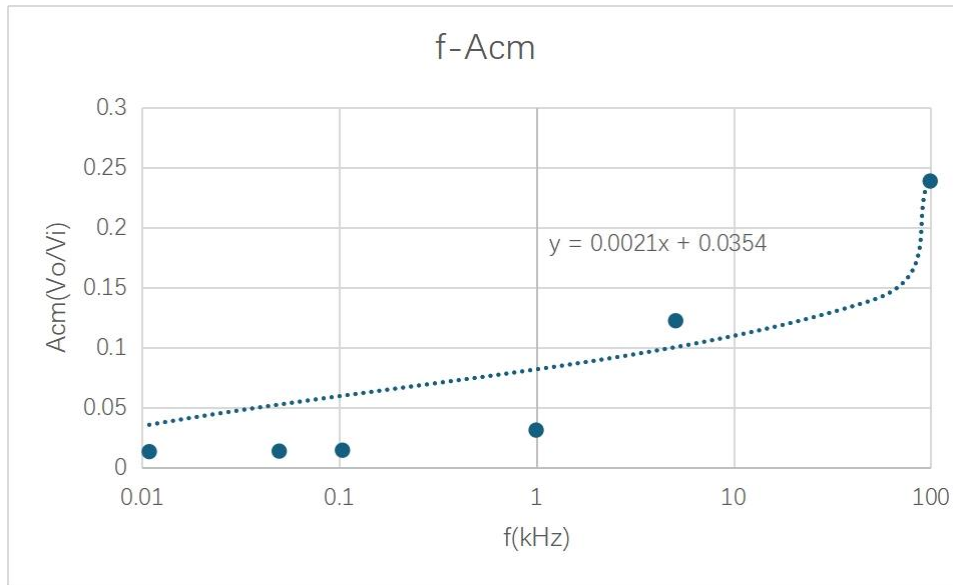


Fig.4. Relationship between the frequency and common mode gain.

d) Testing through the ECG stimulator.

After changing the input source to an ECG stimulator, we can observe the ECG signal from lead I (Original setup: f=50Hz; V_{ipp} = 2mV; BPM = 80) that is being amplified by our instrumentation amplifier, which is shown as Figure 5 and 6 on oscilloscope. We can observe that the peak-to-peak of the amplified ECG signal is 1.38V so that the differential mode gain is $Adm = \frac{V_{out}}{V_{in}} = \frac{1.38}{0.002} = 690$. Also, the heart rate of the ECG waveform can be calculated from the oscilloscope as $Heart\ rate = \frac{1}{7.5 \times 0.1} \times 60 = 80\ bpm$, where 7.5 is the blocks between the 2 “R” points of the ECG waveform, and each block means 0.1s.

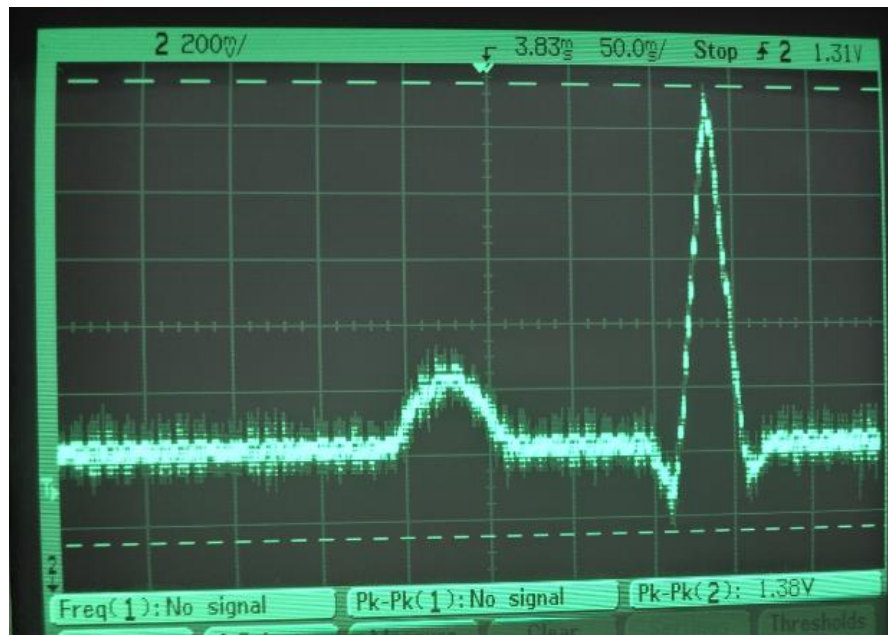


Fig.5. Amplified ECG waveform.

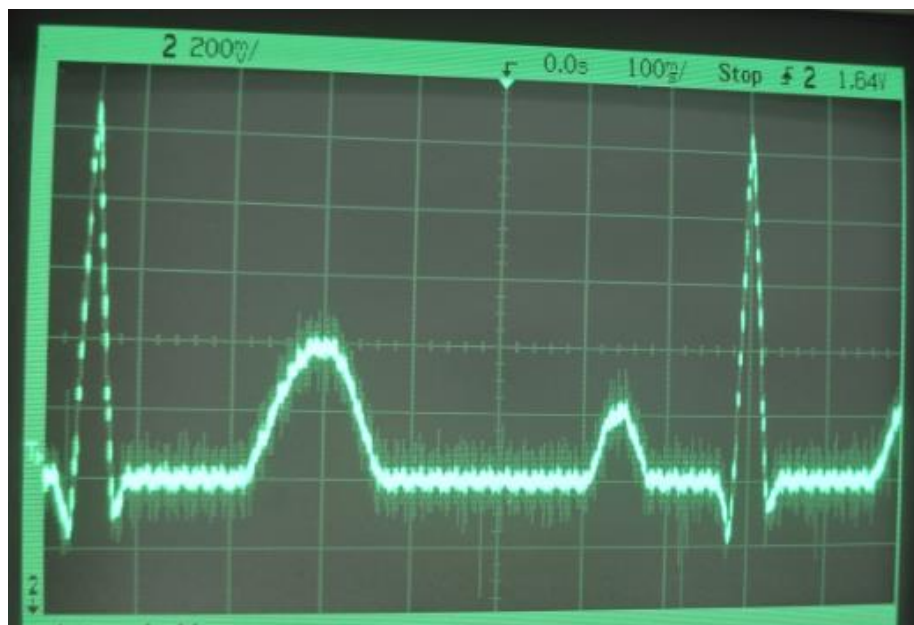


Fig.6. Amplified ECG waveform.

e) Changes of R_{cm} .

After changing the input voltages to ± 6 volt DC source, we can adjust the R_{cm} to 4.89Ω to make the V_{cm} reaches 0.

f) Other pictures of the results are shown in appendix.

2. *A photo of the constructed instrumentation amplifier should be attached.*

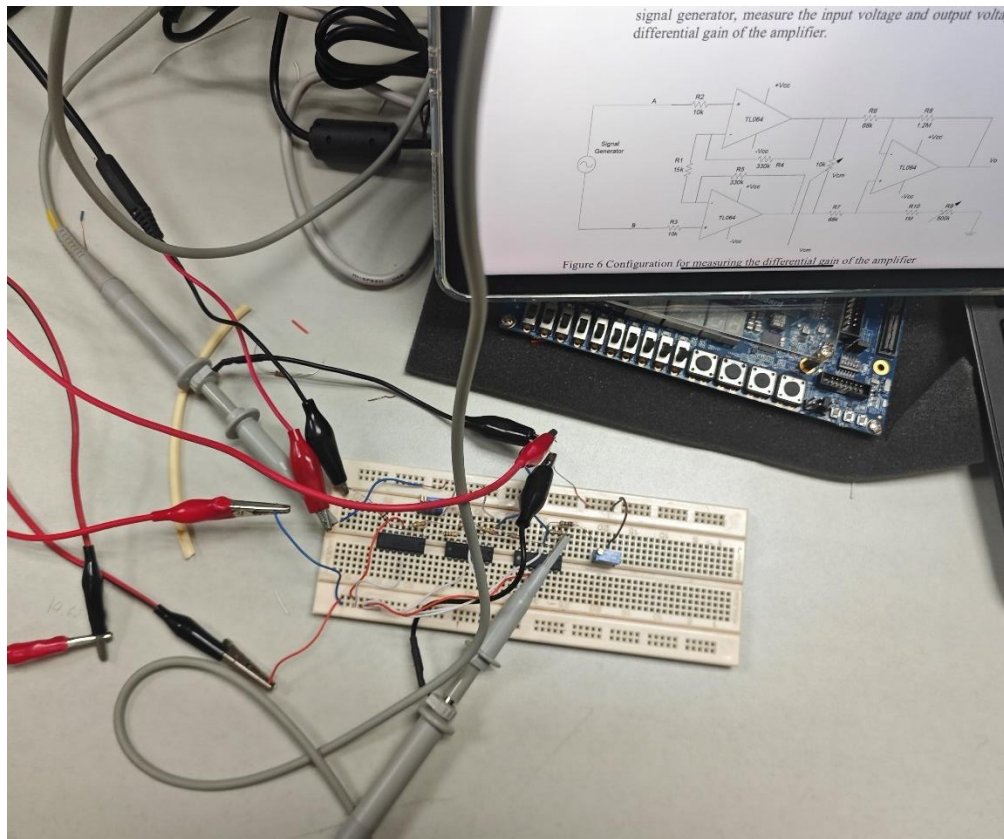


Fig.7 The constructed instrumentation amplifier we use.

3. *What is the CMRR of the amplifier vs. frequency?*

The relationship between CMRR and frequency when $R_9=120k$ are shown as Table 3 and Figure 8.

f(kHz)	0.05	0.1	1	5	100
Adm	752.6221	771.5957	752.6221	692.5388	30.67409
Acm	0.013396	0.014151	0.030943	0.12217	0.238679
CMRR	94.99204	94.73205	87.72029	75.06959	42.17915

Table 3. Data of Instrumentation amplifier at frequency(50-100kHz), $R_9 = 120K\Omega$.

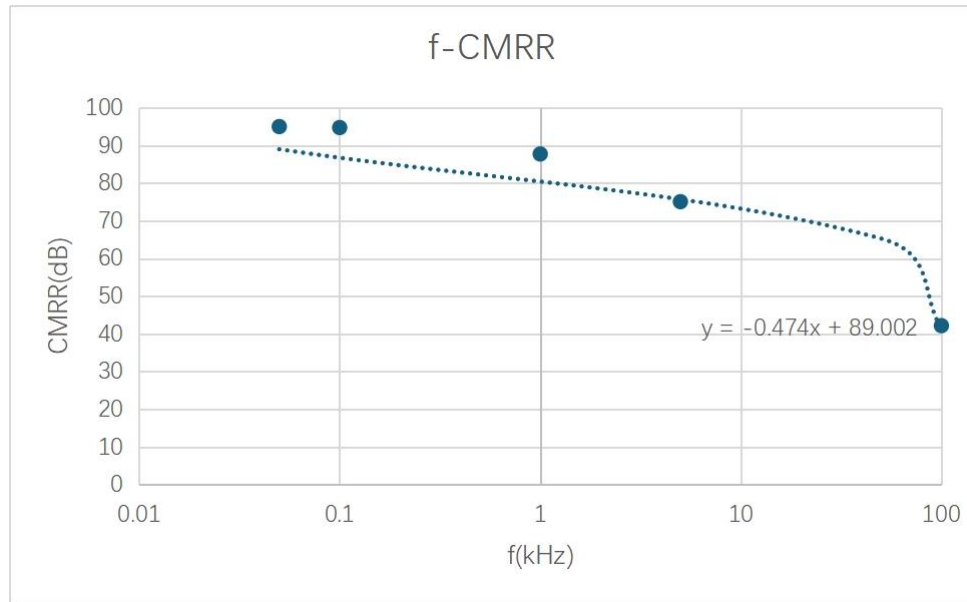


Figure 8. Relationship between CMRR and frequency. (f = 50-100kHz, R9 = 120kΩ)

4. Is R9 important for the differential gain?

As mentioned above, the formula of the differential gain:

$$Gain_{dm} = \left(1 + 2 \frac{R4}{R1}\right) \frac{R8}{R6} = \left(1 + 2 \times \frac{330k}{15k}\right) \times \frac{1.2M}{68k} = 794.1176$$

Therefore, R9 will not influence the differential gain.

5. For an arbitrary value of R9 except the value obtained in part a) of the experiment, what will be the CMRR of the amplifier?

For an arbitrary value of R9=120k, the CMRR of the amplifier

$$CMRR = 20 \log_{10} \left(\frac{A_{dm}}{A_{cm}} \right) = 20 \log_{10} \left(\frac{771.5957}{0.014151} \right) = 94.73205 \text{ dB, where } f = 100\text{Hz}$$

6. Can you comment on the relationship of the resistances of Rcm from each end to the center tap with the common mode voltage obtained?

Rcm provides the bias voltage to the input of the second stage op-amp, which can affect the common mode rejection ability of the instrumentation amplifier. Rcm acts as a variable resistor and the middle pin is connected to the common mode voltage point Vcm and affects the level matching at the input. Adjusting the slide position of Rcm can compensate for the asymmetry of the two first-stage op-amp outputs and reduce the effect of common mode signals. When the resistance of Rcm to both ends is equal (10k/2 = 5k), the common mode voltage at the inputs of the second stage op-amps is divided equally. Better

common mode rejection is achieved.

7. *Comment on each result that you obtained in the experiment.*

a) Differential mode analysis.

The oscilloscope we used are quite old, it can't show the waveform when frequency lower than 50Hz and Higher than 100kHz. Some part of A_{dm} we can't analysis. We can find the differential mode gain is around 760 V/V that is similar to the calculation result of our prelab equals to 794 V/V. Moreover, as we see from Figure 2, the cutoff frequency is around 10k Hz and as an ECG amplifier, this design is suitable for the low frequency signals.

b) Common mode analysis.

For $R_9 = 500k\Omega$, the A_{cm} is not small enough to depress the common mode signal.

c) Improvement of CMRR

After we tuning the R_9 to $120k\Omega$, the performance of common mode signal depress is better in $f=50Hz$.

d) Testing through the ECG stimulator.

After we connect the ECG stimulator, we can clearly see the amplified ECG signal. It means that the common mode signal depress ability of our amplifier is good, also the gain of amplifier is up to standard. We think that at least it can be used.

e) Changes of R_{cm} .

The V_{cm} reach zero when $R_{cm} = 4.89\Omega$ in our experiment. It is different to the ideal value. We think it is because some instrumental error like resistance error or possible input bias current in the amplifier components and then cause asymmetry.

VII. CONCLUSION

Through this experiment, we delved into various aspects of biomedical signal measurement, from electrocardiogram (ECG) to electromyogram (EMG) to blood pressure measurement. These experiments not only consolidated the theoretical knowledge learnt in class, but also enhanced our understanding and ability to apply practical exercises. The results of each experiment provided us with valuable lessons that helped us better understand the properties of biomedical signals and their importance in clinical applications. Looking ahead, we will continue to explore this field to further enhance our skills and knowledge.

VIII. APPENDIX

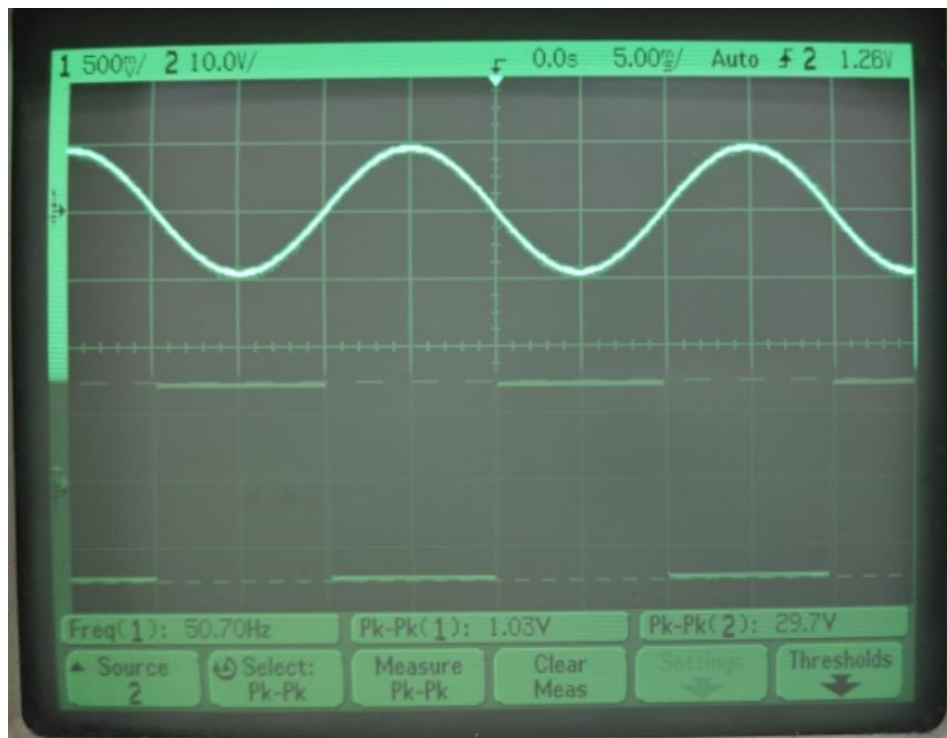


Fig.a. Clipping waveform

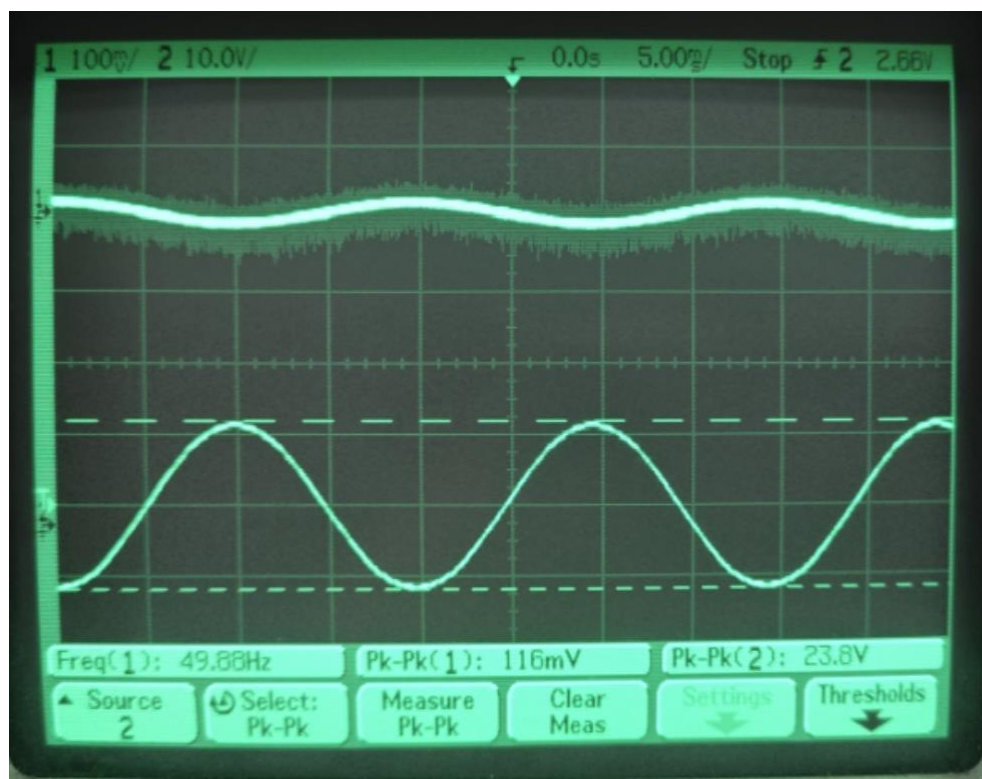


Fig.b. Differential mode waveform (50Hz)

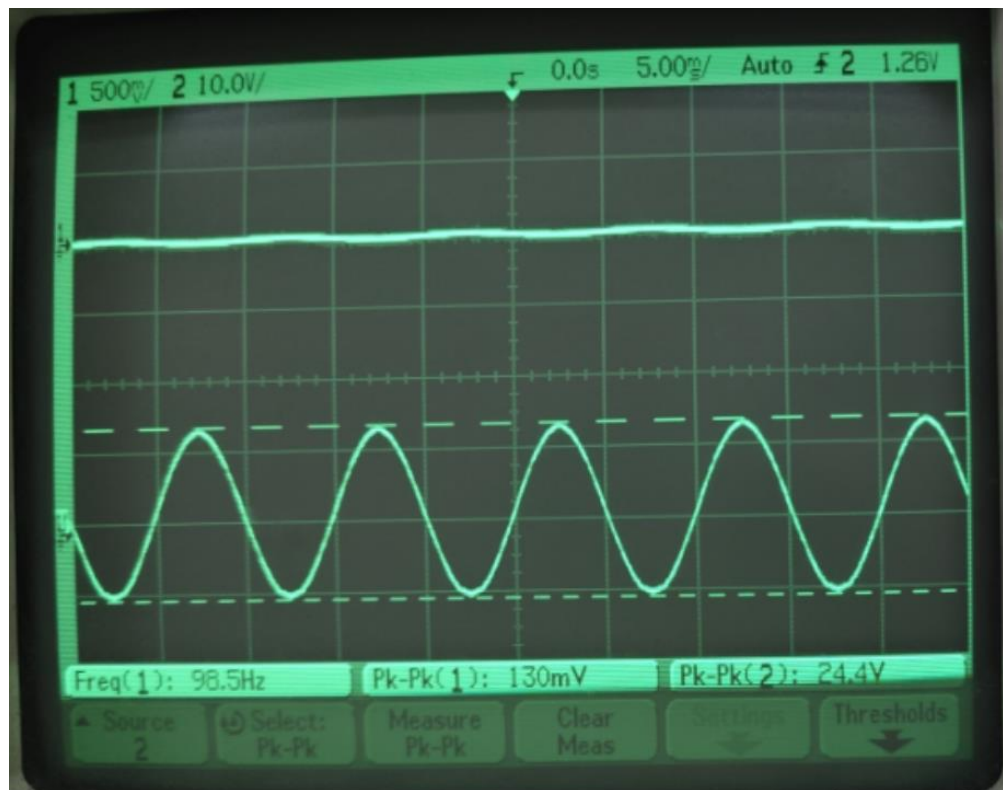


Fig.c. Differential mode waveform (100Hz)

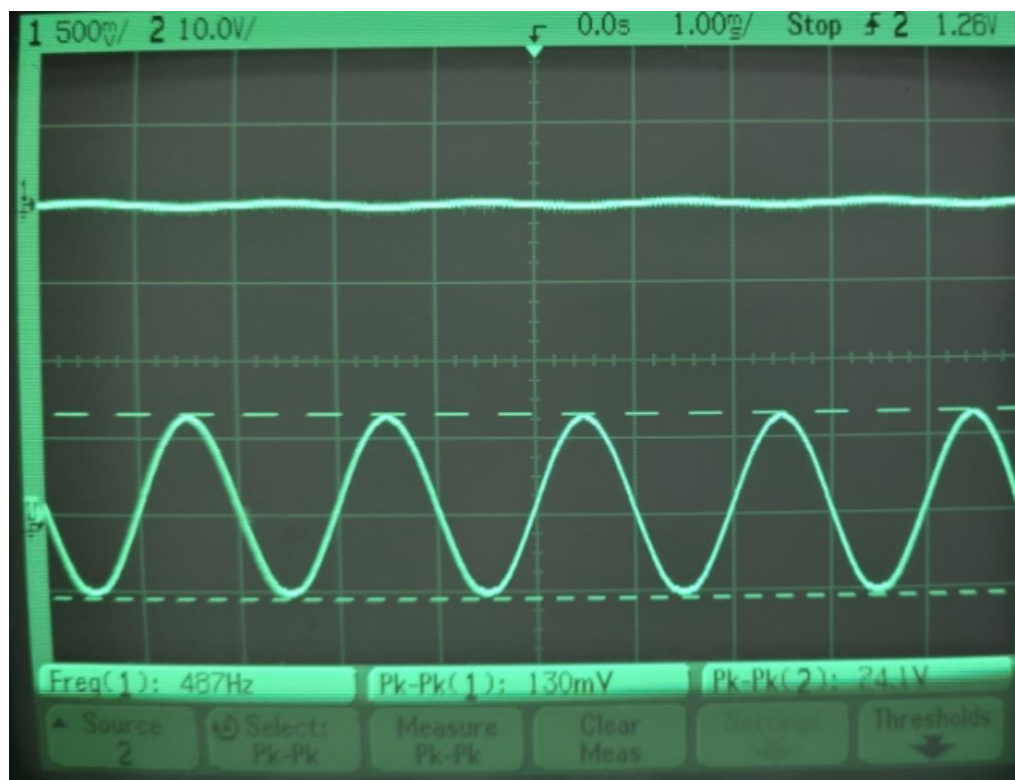


Fig.d. Differential mode waveform (500Hz)

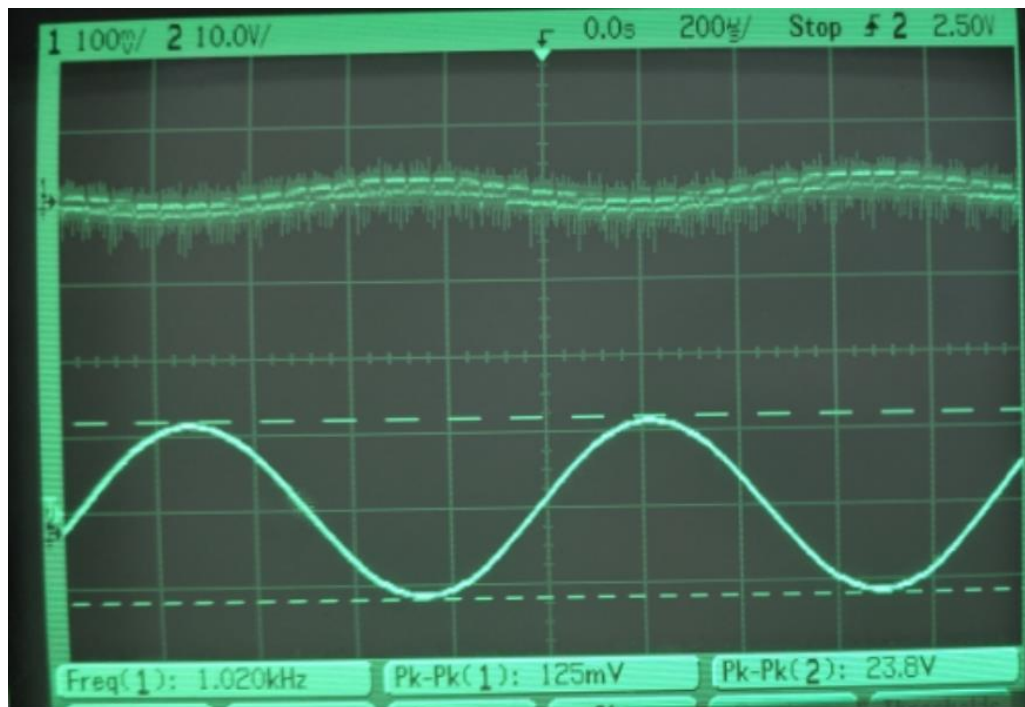


Fig.e. Differential mode waveform (1kHz)



Fig.f. Differential mode waveform (5kHz)

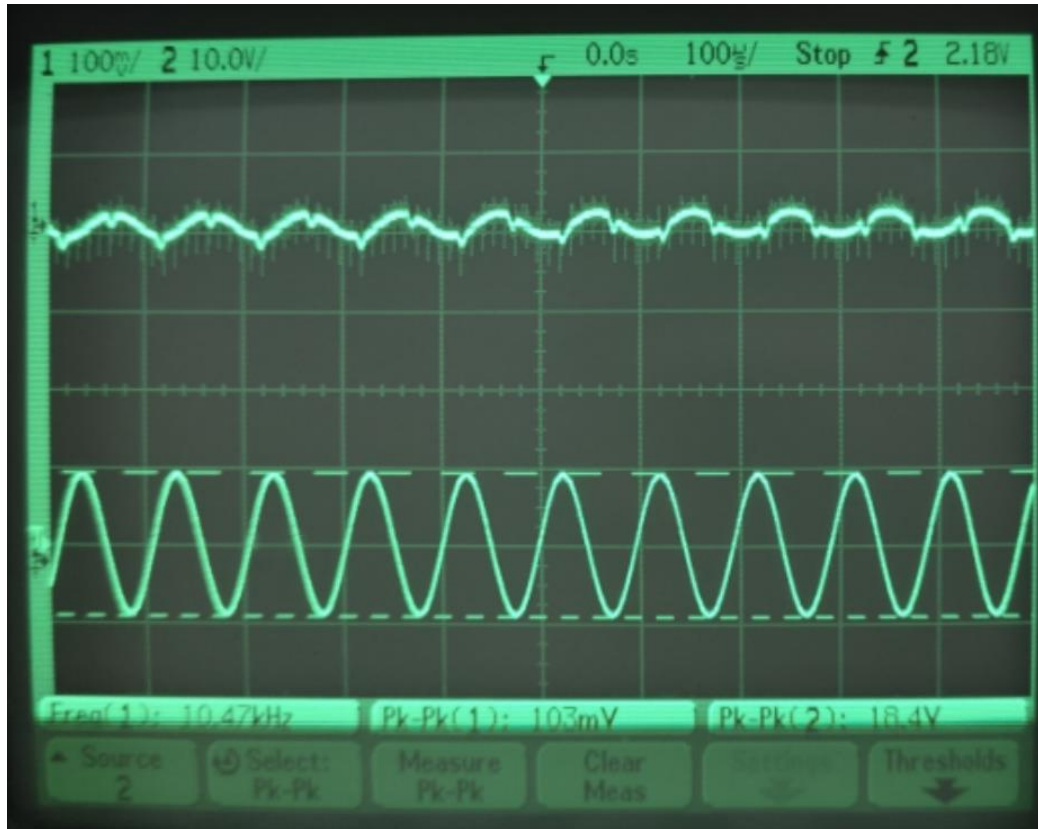


Fig.g. Differential mode waveform (10kHz)

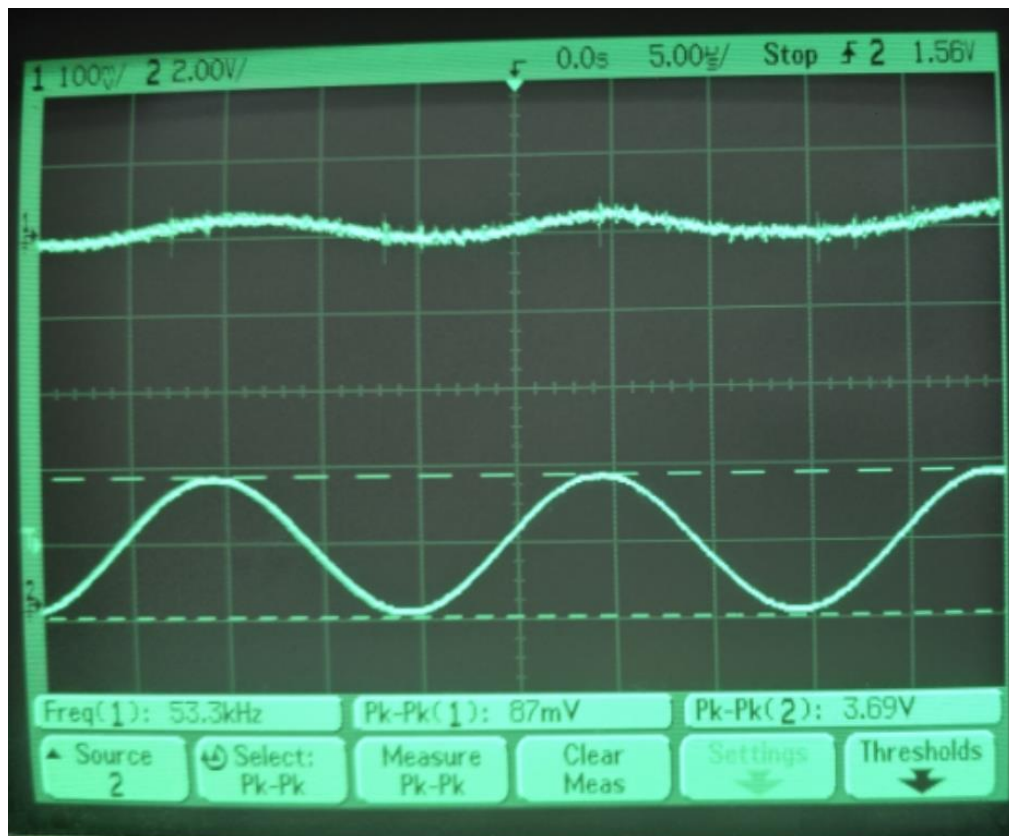


Fig.h. Differential mode waveform (50kHz)

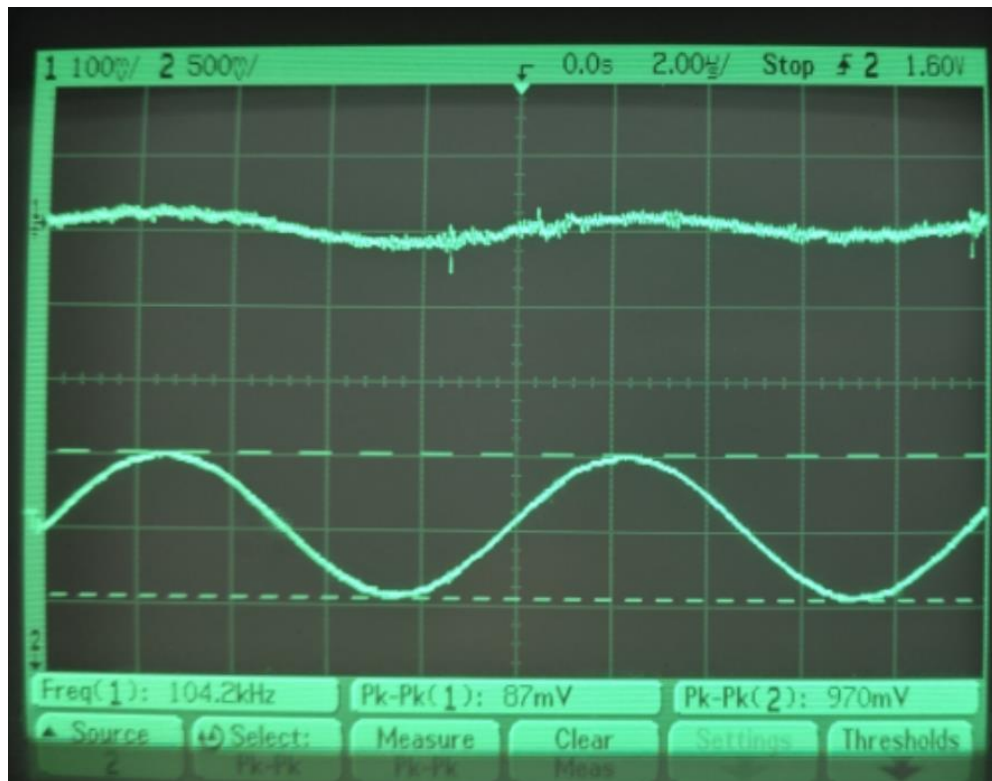


Fig.i. Differential mode waveform (100kHz)

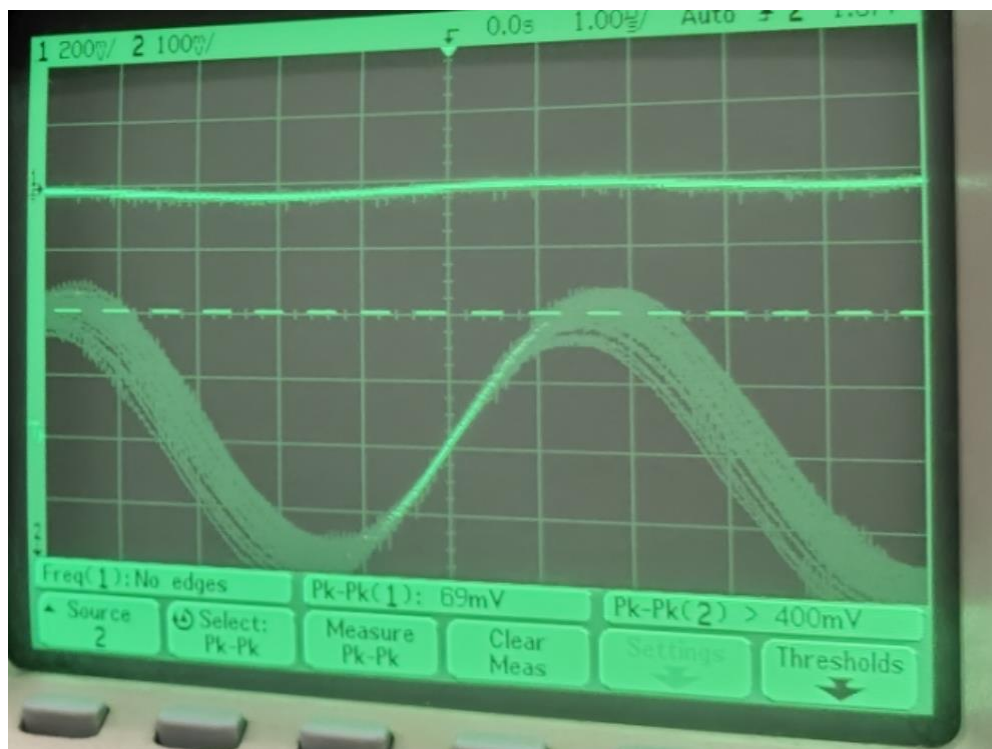


Fig.j. Differential mode waveform (200kHz)

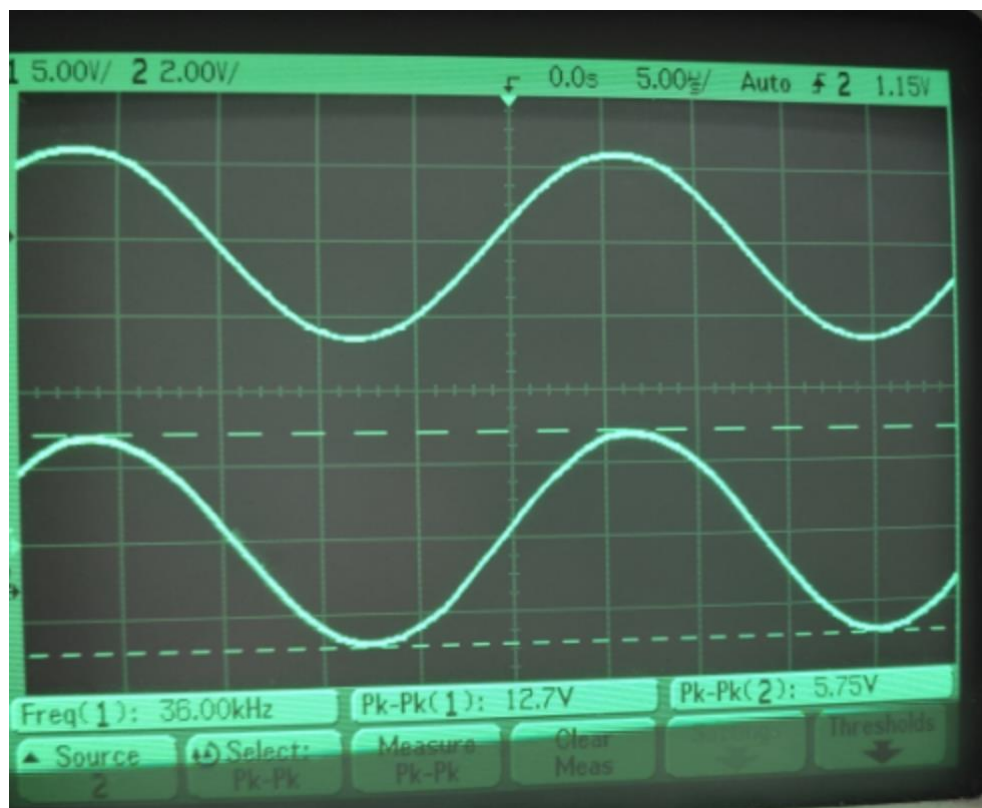


Fig.k. Common mode waveform ($f=36\text{kHz}$, $R_9 = 500\text{k}\Omega$)

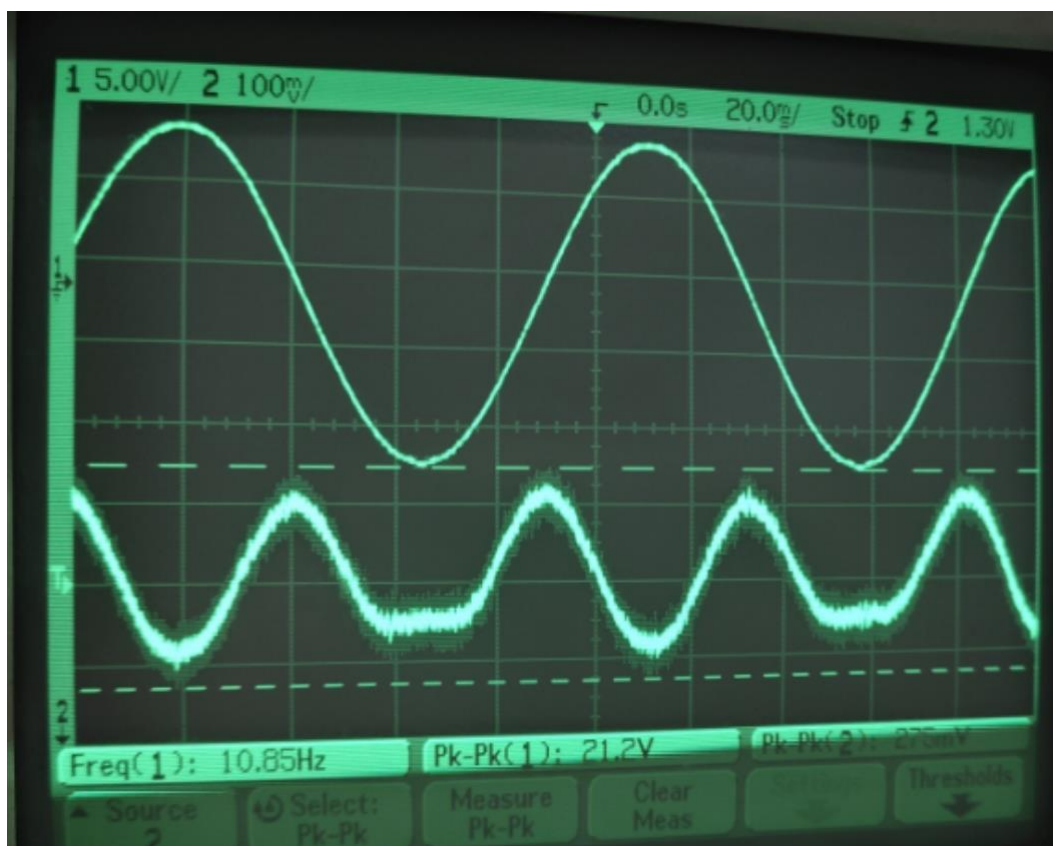


Fig.l. Common mode waveform ($f=10\text{Hz}$, $R_9 = 120\text{k}\Omega$)

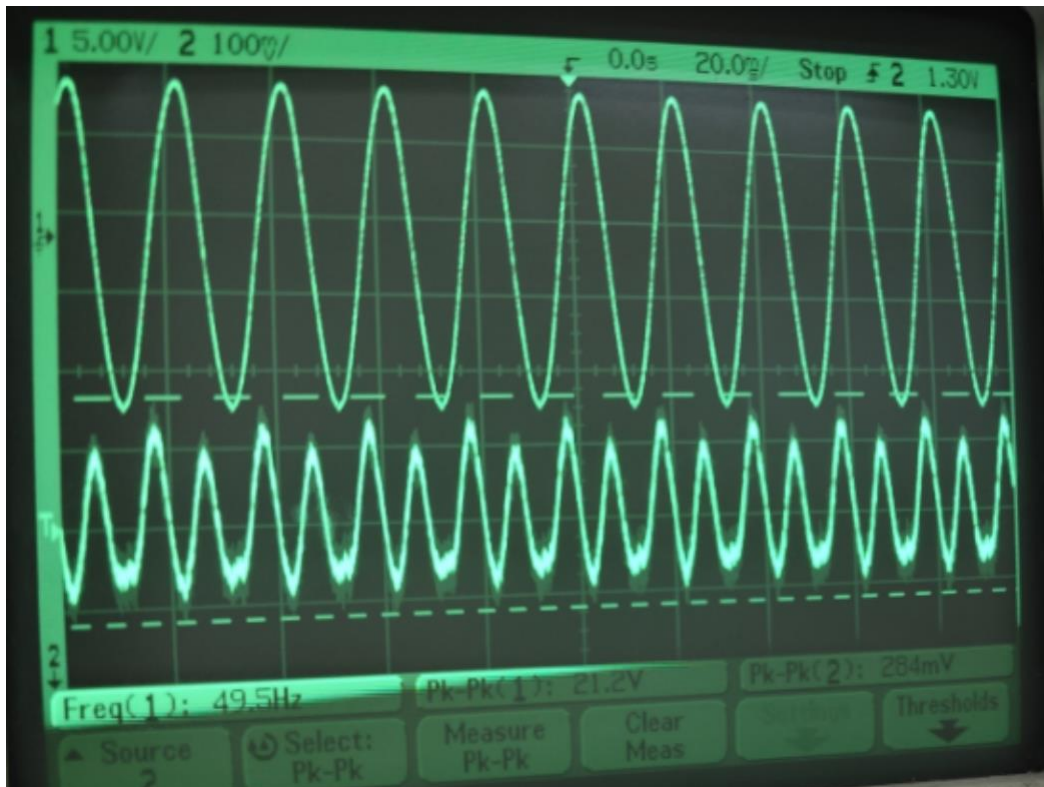


Fig.m. Common mode waveform ($f=50\text{Hz}$, $R_9 = 120\text{k}\Omega$)

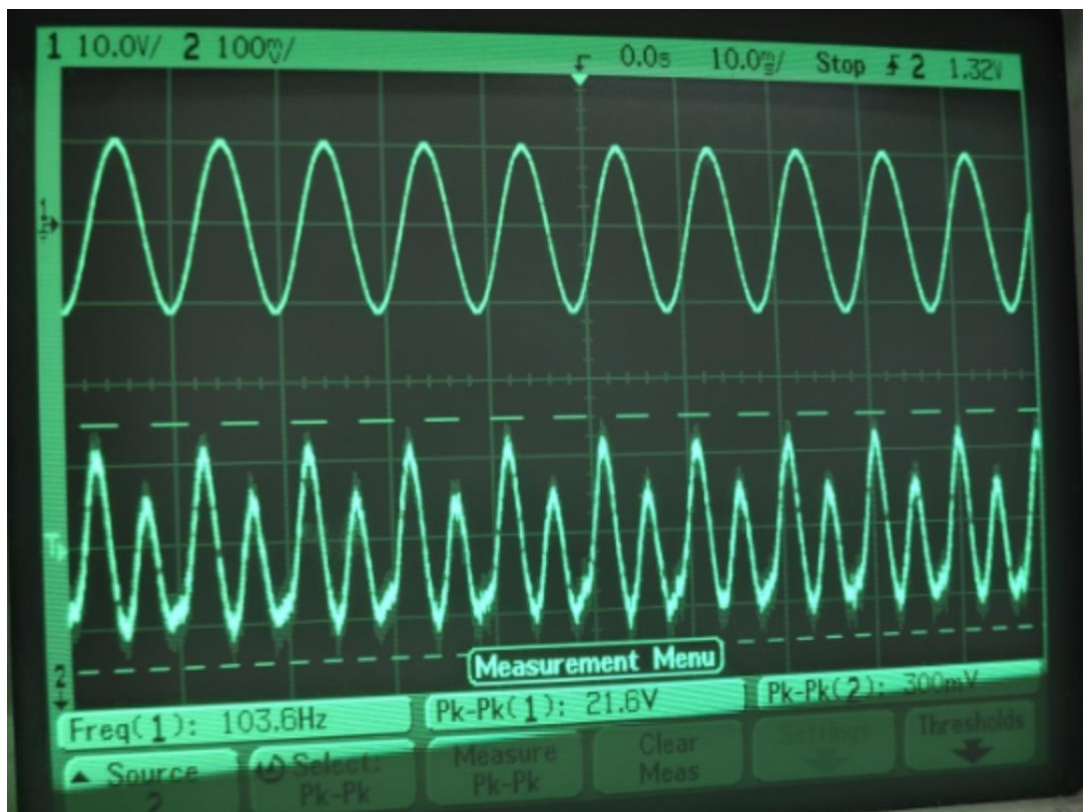


Fig.n. Common mode waveform ($f=100\text{Hz}$, $R_9 = 120\text{k}\Omega$)

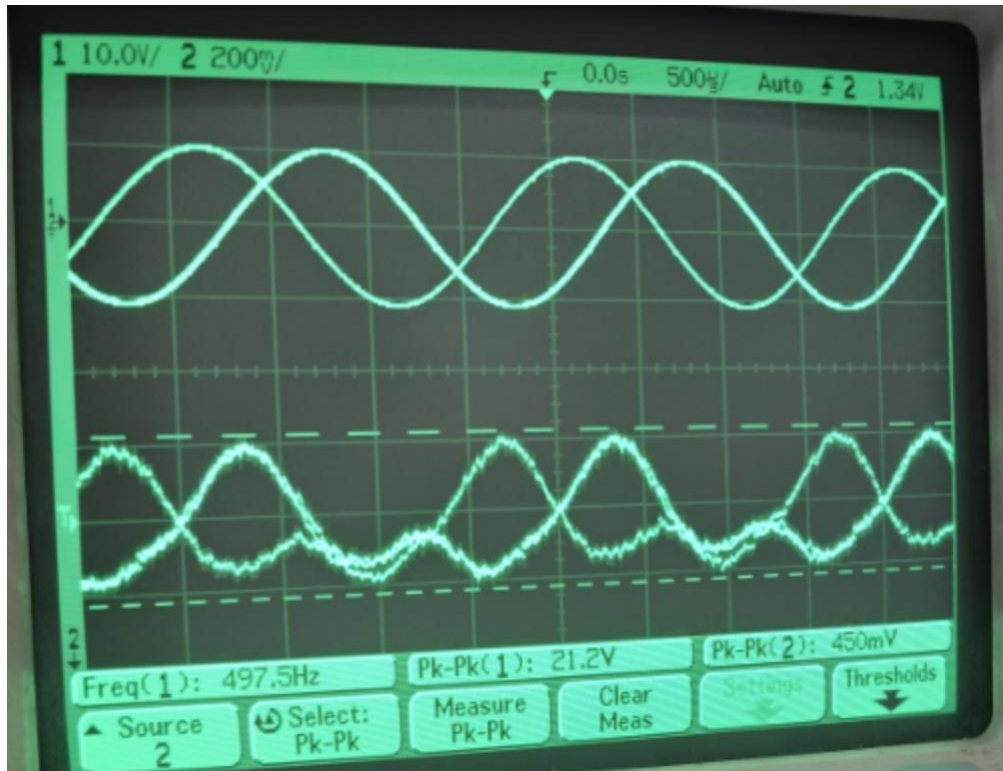


Fig.o. Common mode waveform ($f=500\text{Hz}$, $R_9 = 120\text{k}\Omega$)

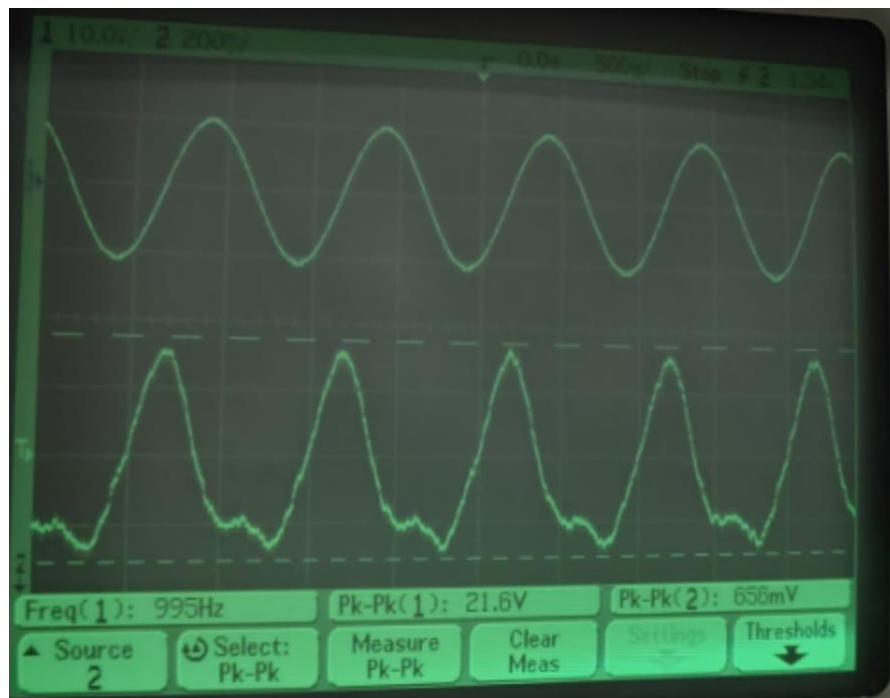


Fig.p. Common mode waveform ($f=1\text{kHz}$, $R_9 = 120\text{k}\Omega$)

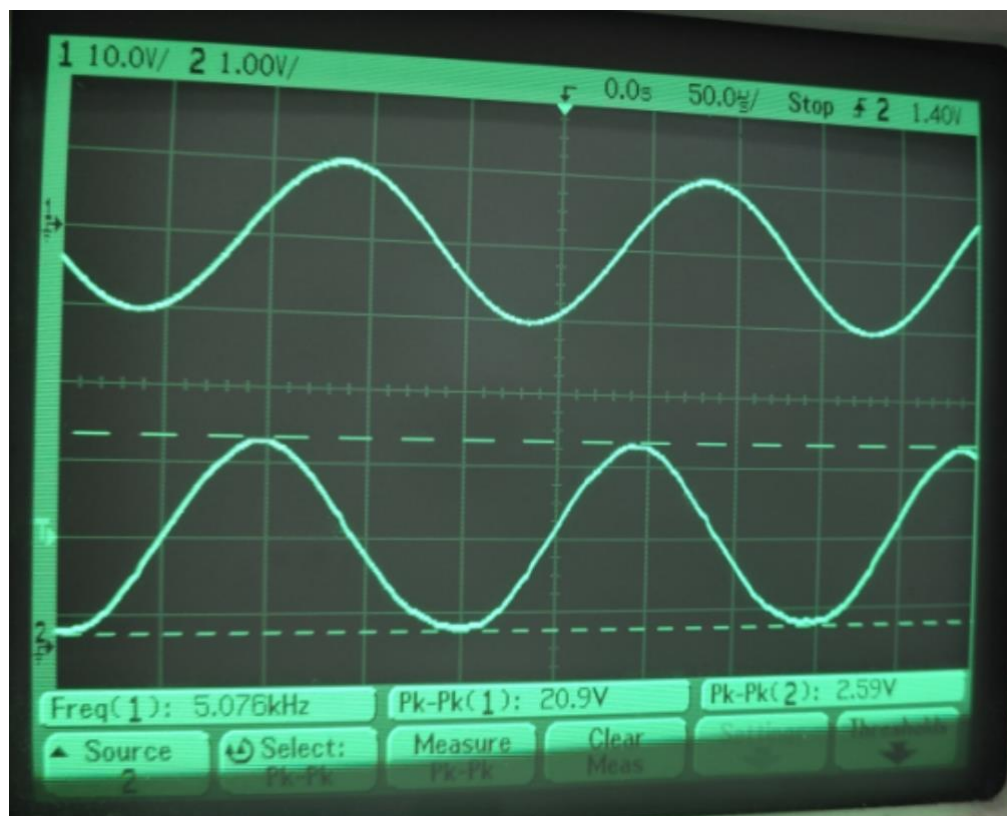


Fig.q. Common mode waveform ($f=5\text{kHz}$, $R_9 = 120\text{k}\Omega$)

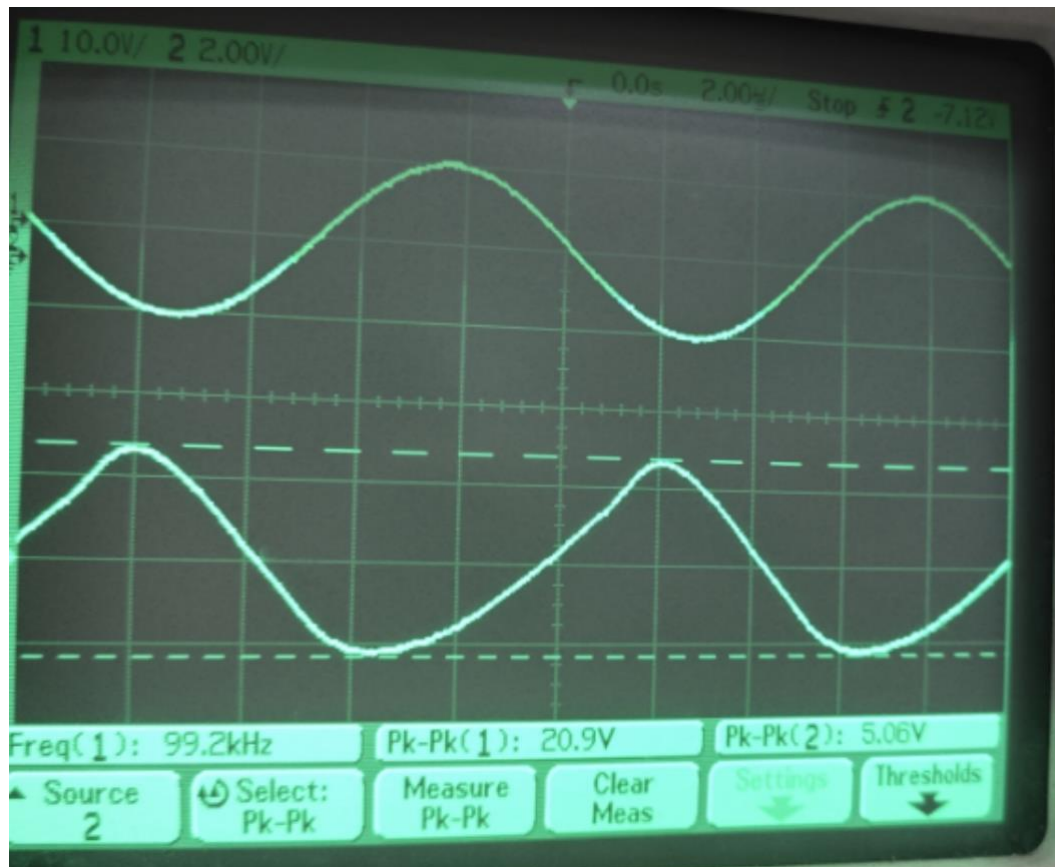


Fig.r. Common mode waveform ($f=100\text{kHz}$, $R_9 = 120\text{k}\Omega$)

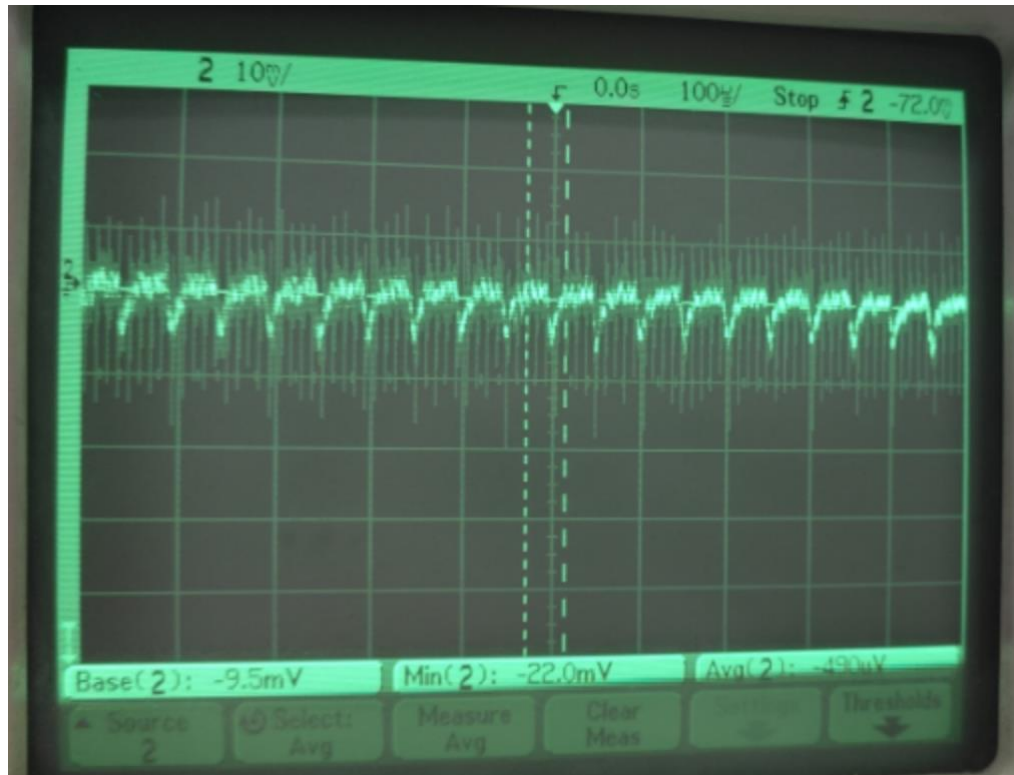


Fig.s. Common mode waveform ($f=100\text{kHz}$, $R_{cm} = 4.8\text{k}\Omega$, $R_9 = 120\text{k}\Omega$)